

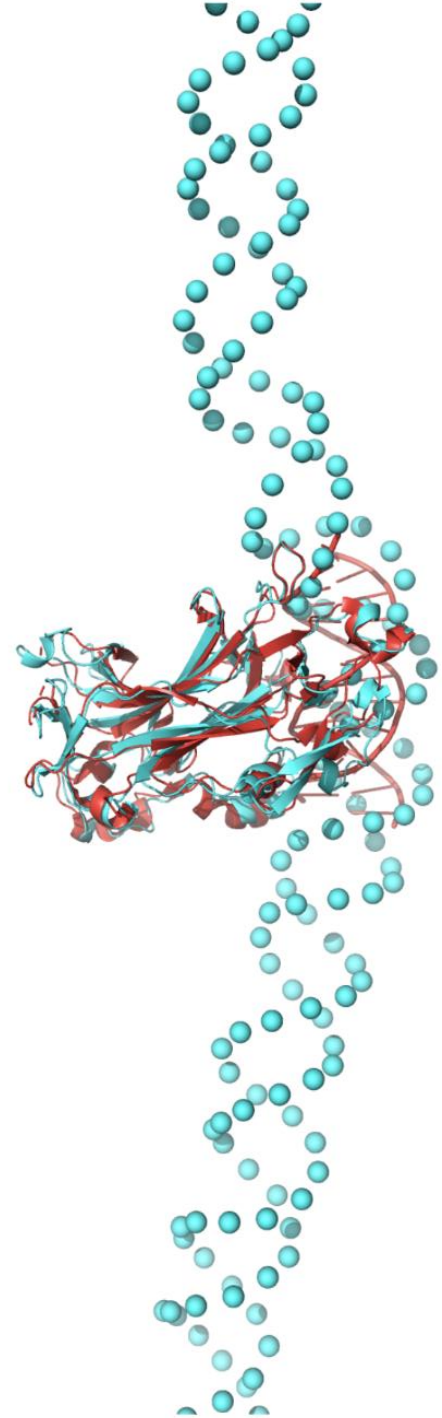


The associative memory, water mediated, structure and energy model (AWSEM)

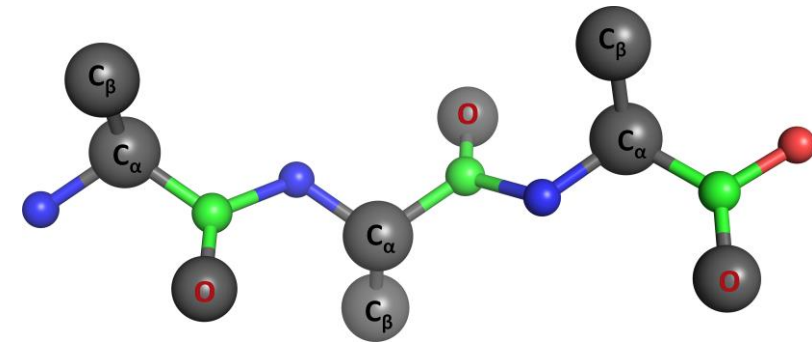
Carlos Bueno

AWSEM

- General protein forcefield.
- Uses:
 - Protein structure prediction.
 - Protein folding and misfolding pathways.
 - Protein binding and aggregation
 - Association with DNA, RNA and membranes.
 - Frustration and stability prediction.



The AWSEM Forcefield



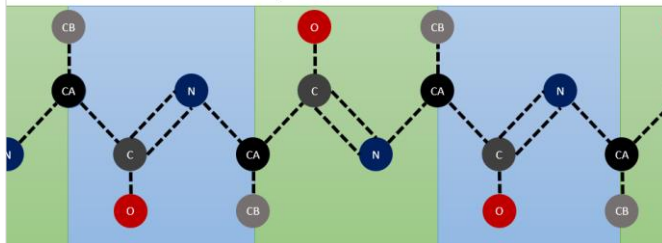
$$V_{AWSEM} = V_{backbone} + V_{contact} + V_{Hbonds} + V_{memory} + V_{extra}$$

- $V_{backbone}$: Each residue is modeled by 3 particles. Implicit representations of the water, hydrogens and sidechains.
- $V_{contact}$: Many-body terms define the local residue interactions
- V_{Hbonds} : Bias proteins conformation to form alpha helices and beta sheets by implicit modeling of backbone hydrogen bonding
- V_{memory} : Bias protein secondary structure to conformations found using bioinformatics tools, atomic models or other structural data.
- V_{extra} : Can be supplemented with additional terms like evolutionary restraints, electrostatics, etc.

$$V_{AWSEM} = V_{backbone} + V_{contact} + V_{Hbonds} + V_{memory} + V_{extra}$$

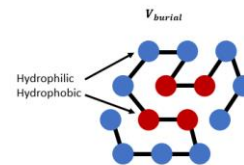
Backbone potential ($V_{backbone}$)

$$V_{backbone} = V_{connectivity} + V_{chain} + V_{chi} + V_{Rama} + V_{Exclusion}$$

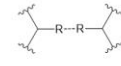


Contact and Burial potentials ($V_{contact}$)

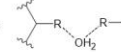
$$V_{contact} = V_{burial} + V_{direct} + V_{water-mediated} + V_{protein-mediated}$$



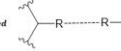
V_{direct}



$V_{water-mediated}$



$V_{protein-mediated}$



Hydrogen bonding terms

Assumption: Secondary structure forms as a cooperative process that depends on distance, orientation, and amino acid nature

$$V_{HB} = V_{\beta} + V_{P-AP} + V_{helical}$$



Associative memory term

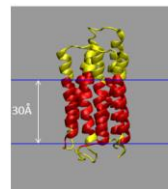
Assumption: The details absent from the AWSEM model can be extracted from known structures.

$$V_{AM} = -\lambda_{AM} \sum_m \omega_m \sum_{ij} \gamma_{ij} \exp \left[-\frac{(r_{ij} - r_{ij}^m)^2}{2\sigma_{ij}^2} \right]$$

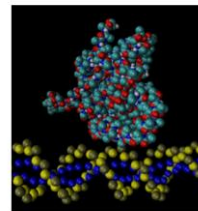
- Range: Secondary to super-secondary structure ($|i - j| \leq 12$)
- Parameters: Memories can be obtained by using bioinformatic searches to obtain global or local alignments or by simulating fragments using all-atom simulations

Extra terms

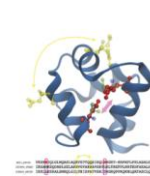
Membrane Potential



Open3SPN2 DNA model

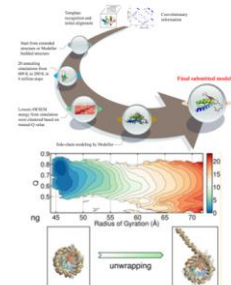


Evolutionary restraints



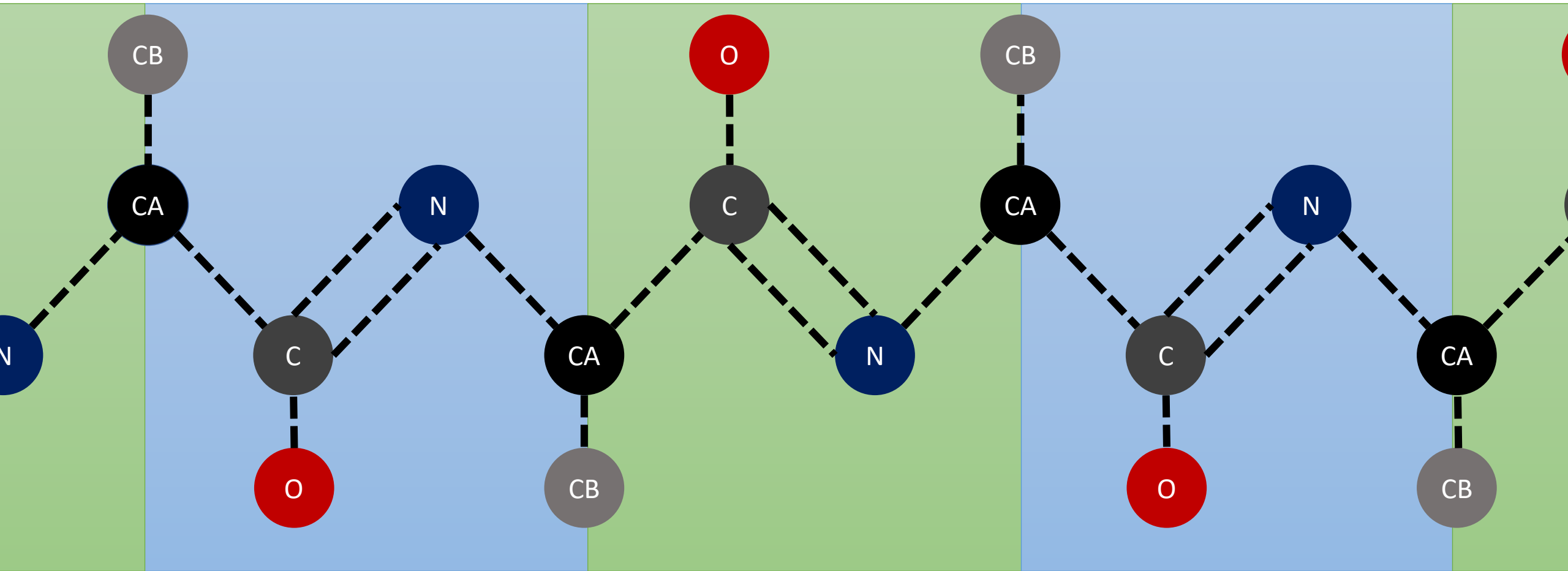
Some applications

- Protein structure prediction.
- Protein folding and misfolding pathways.
- Protein binding and aggregation
- Association with DNA, RNA and membranes.
- Frustration and stability prediction.

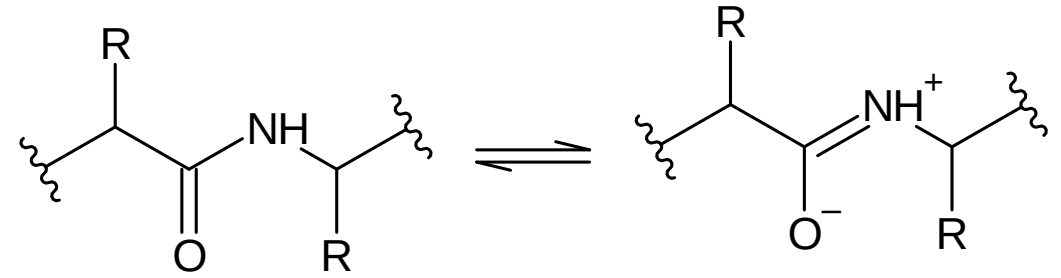


Backbone potential ($V_{backbone}$)

$$V_{backbone} = V_{connectivity} + V_{chain} + V_{chi} + V_{Rama} + V_{Exclusion}$$



Backbone potential

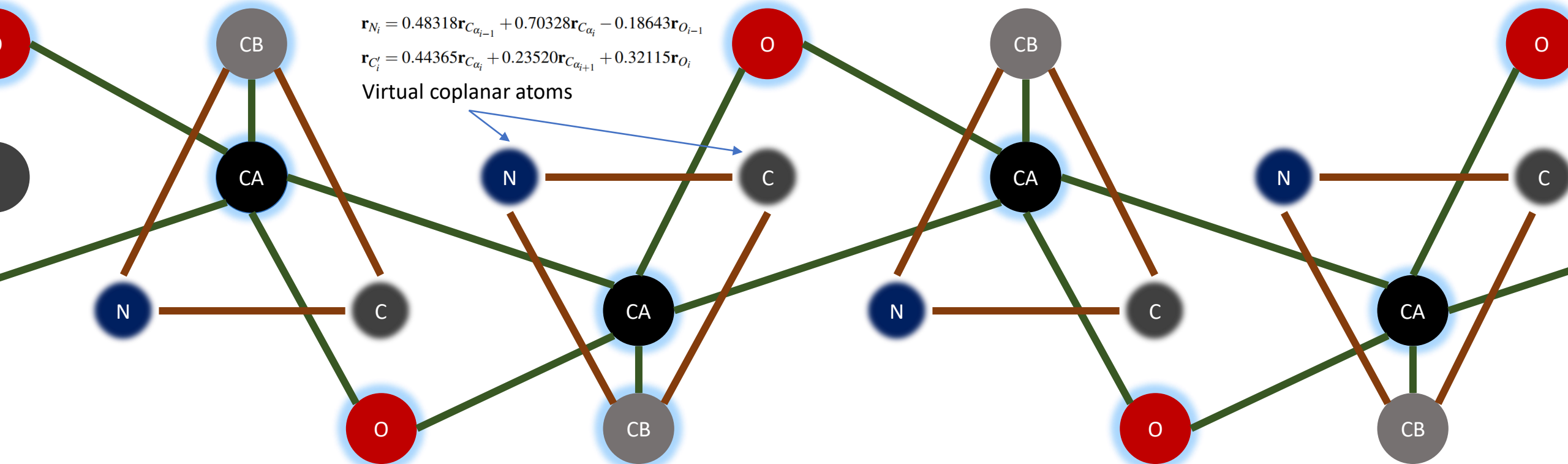


$$V_{backbone} = V_{connectivity} + V_{chain} + V_{chi} + V_{Rama} + V_{Exclusion}$$

Assumption: The peptide bond does not rotate and falls in a plane

$$V_{connectivity} = \sum \lambda_{con} (r_{ij} - r_b)^2$$

$$V_{chain} = \sum \lambda_{chain} (r_{ij} - r_b)^2$$

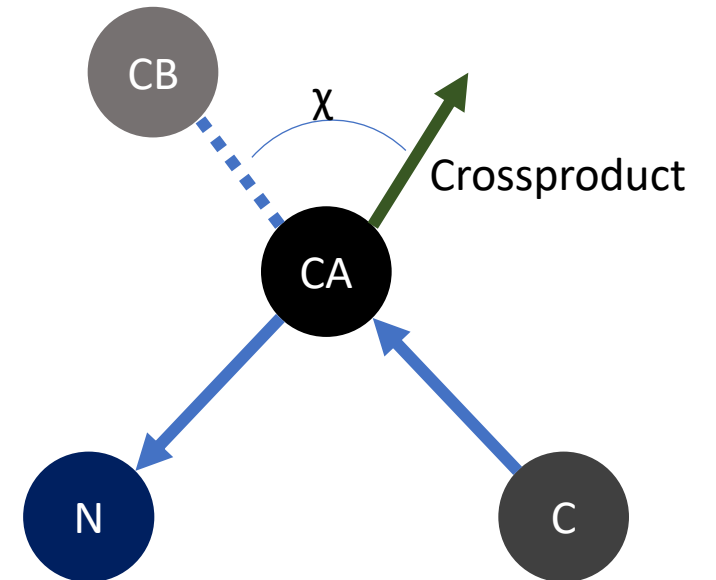


Backbone potential

$$V_{backbone} = V_{connectivity} + V_{chain} + V_{chi} + V_{Rama} + V_{Exclusion}$$

Assumption: The Ca has a tetrahedral shape with a fixed stereochemistry (L-aminoacid)

$$V_{\chi} = \lambda_{\chi} \sum (\chi_i - \chi_0)^2$$
$$\chi_i = \frac{(\mathbf{r}_{C'_i C_{\alpha_i}} \times \mathbf{r}_{C_{\alpha_i} N_i}) \cdot \mathbf{r}_{C_{\alpha_i} C_{\beta_i}}}{|\mathbf{r}_{C'_i C_{\alpha_i}} \times \mathbf{r}_{C_{\alpha_i} N_i}| \cdot |\mathbf{r}_{C_{\alpha_i} C_{\beta_i}}|}$$

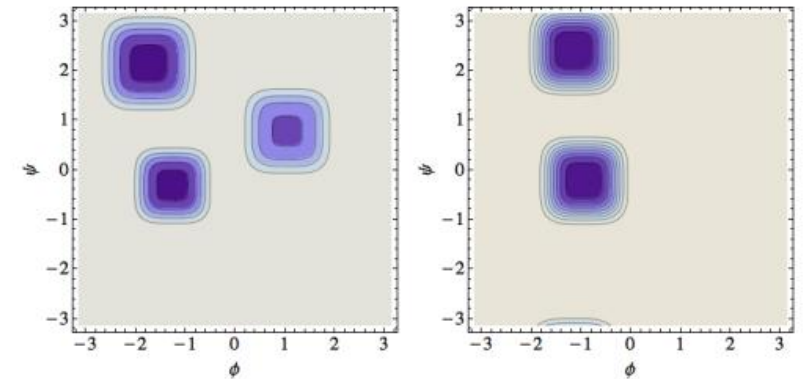
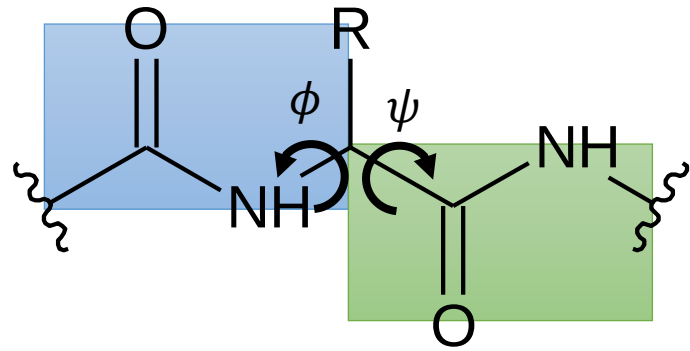


Backbone potential

$$V_{backbone} = V_{connectivity} + V_{chain} + V_{chi} + V_{Rama} + V_{Exclusion}$$

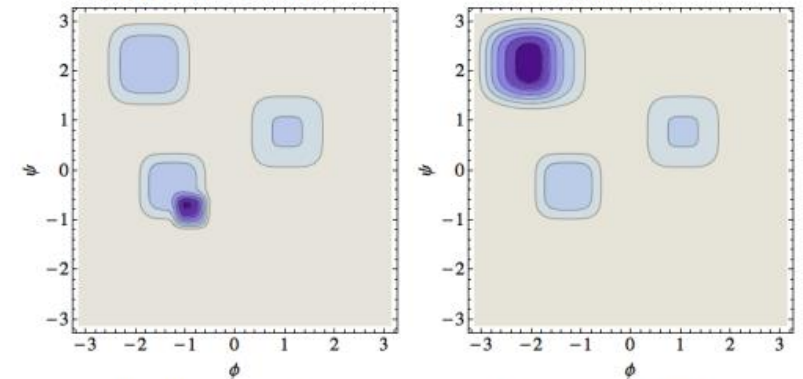
Assumption: Some dihedral angles are favored

$$V_{rama} = -\lambda_{rama} \sum_{i=2}^{N-1} \sum_j W_j e^{-\sigma_j (\omega_{\phi_j} (\cos(\phi_i - \phi_j^0) - 1)^2 + \omega_{\psi_j} (\cos(\psi_i - \psi_j^0) - 1)^2)}$$



(a) All residues except proline

(b) Proline



(c) Biased toward alpha helix

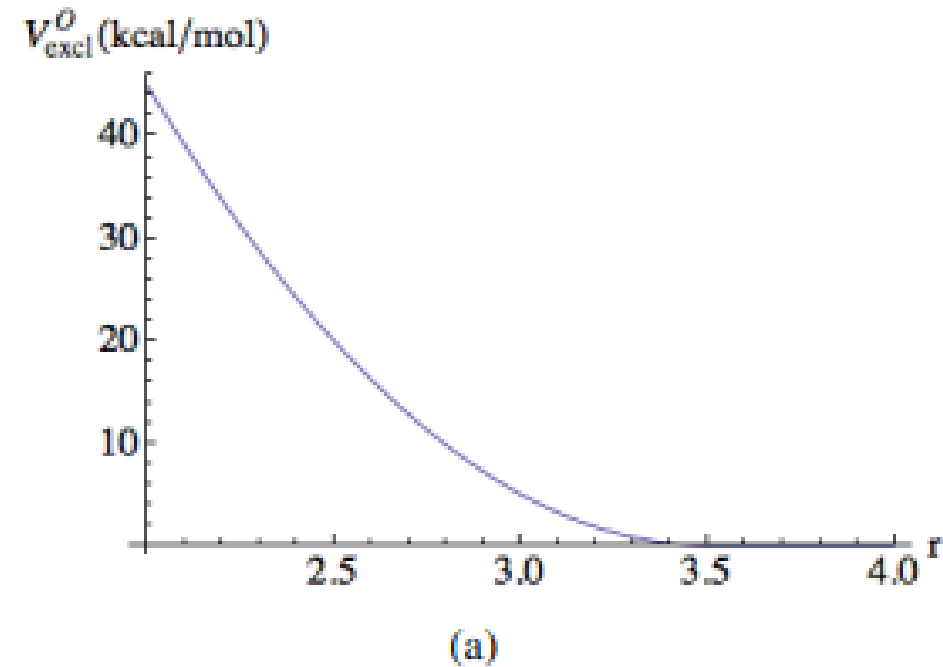
(d) Biased toward beta sheet

Backbone potential

$$V_{backbone} = V_{connectivity} + V_{chain} + V_{chi} + V_{Rama} + V_{Exclusion}$$

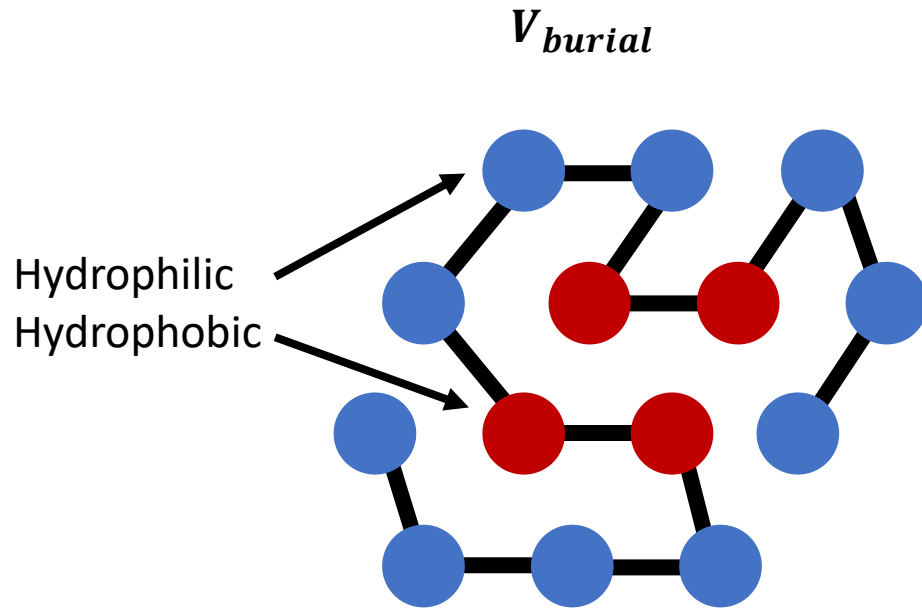
Assumption: Residues don't overlap each other.

$$V_{Exclusion} = \lambda_{excl}(r_{ij} - r_{ex})^2 \quad r_{ij} < r_{ex}$$

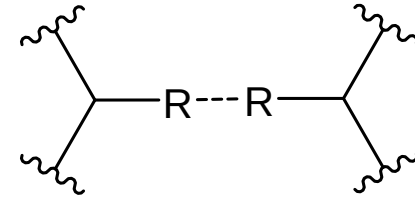


Contact and Burial potentials ($V_{contact}$)

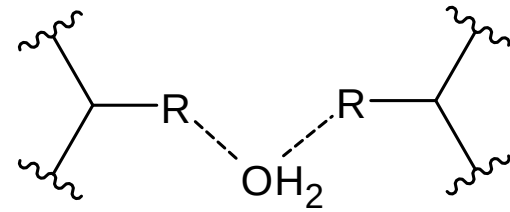
$$V_{contact} = V_{burial} + V_{direct} + V_{water-mediated} + V_{protein-mediated}$$



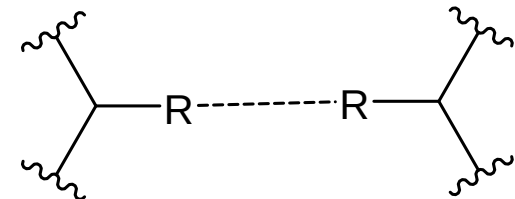
V_{direct}



$V_{water-mediated}$



$V_{protein-mediated}$



Burial term

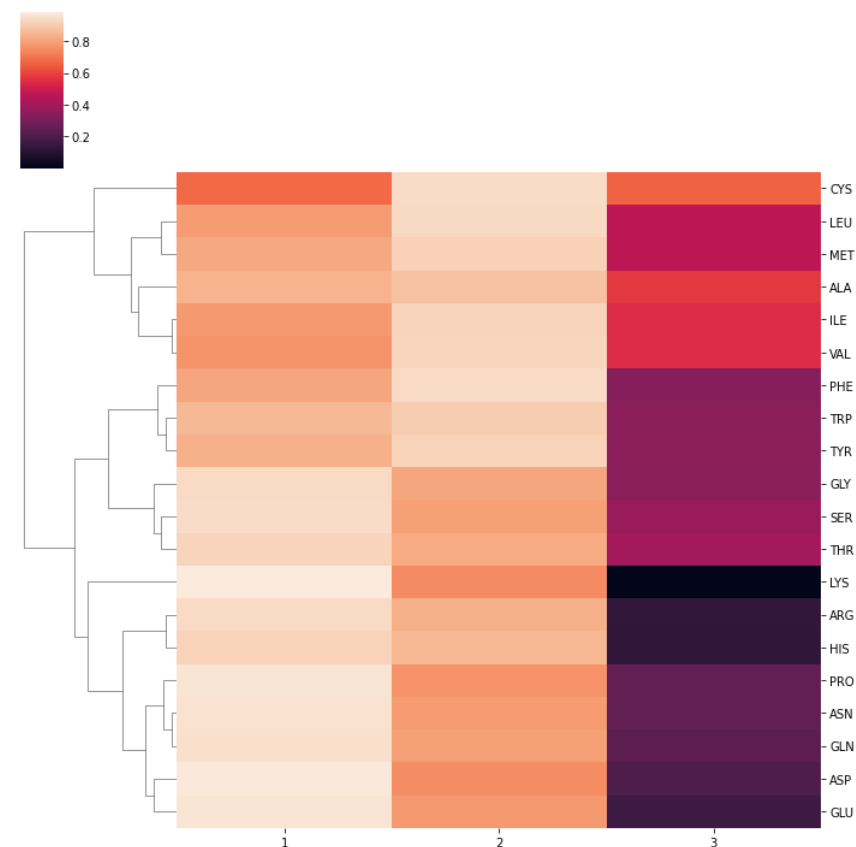
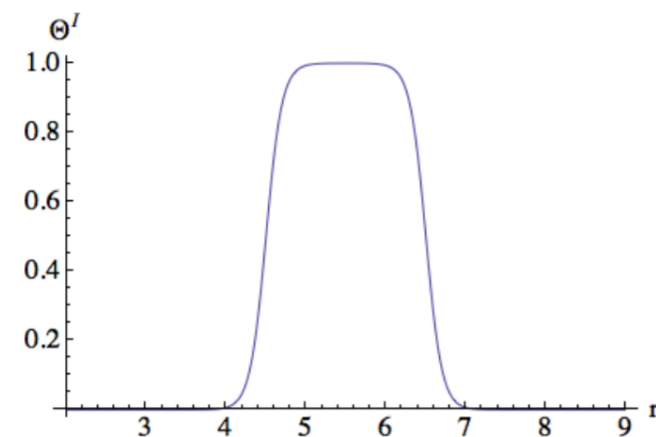
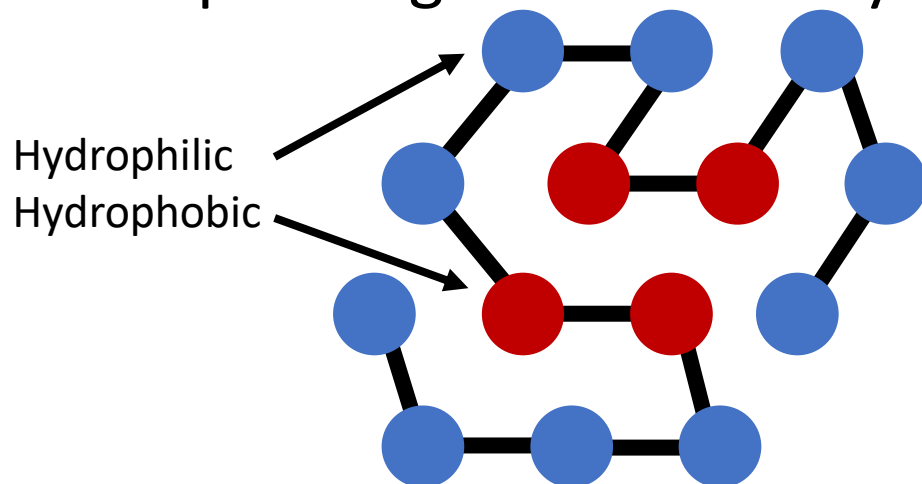
Assumption: Hydrophobic residues tend to be surrounded by other residues

$$\rho_i = \sum_{j=1}^N \Theta_{ij}^I \quad \leftarrow \text{Local protein density a.k.a. "Density"}$$

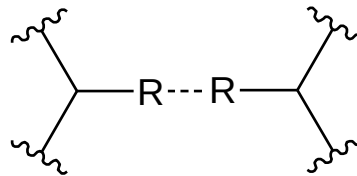
V_{burial}

$$V_{burial} = -\frac{1}{2} \lambda_{burial} \sum_{i=1}^N \sum_{\mu=1}^3 \gamma_{burial}(a_i, \rho_i) \left(\tanh[\eta(\rho_i - \rho_{min}^{\mu})] + \tanh[\eta(\rho_{max}^{\mu} - \rho_i)] \right)$$

- 3 Wells depending on the density of the protein

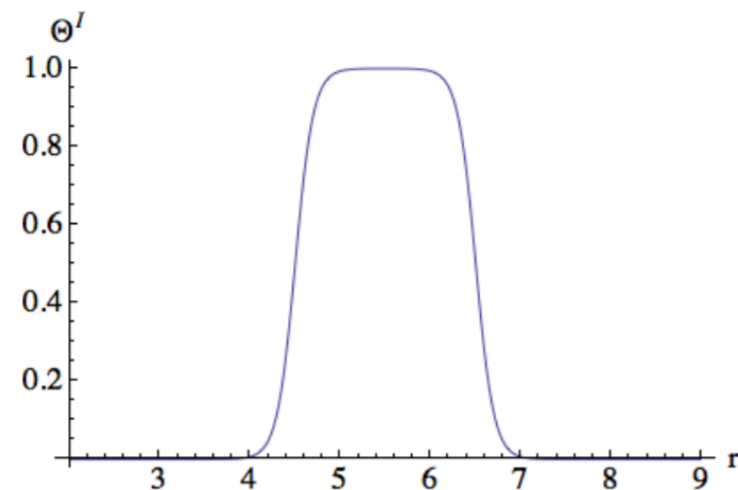


Direct Interactions

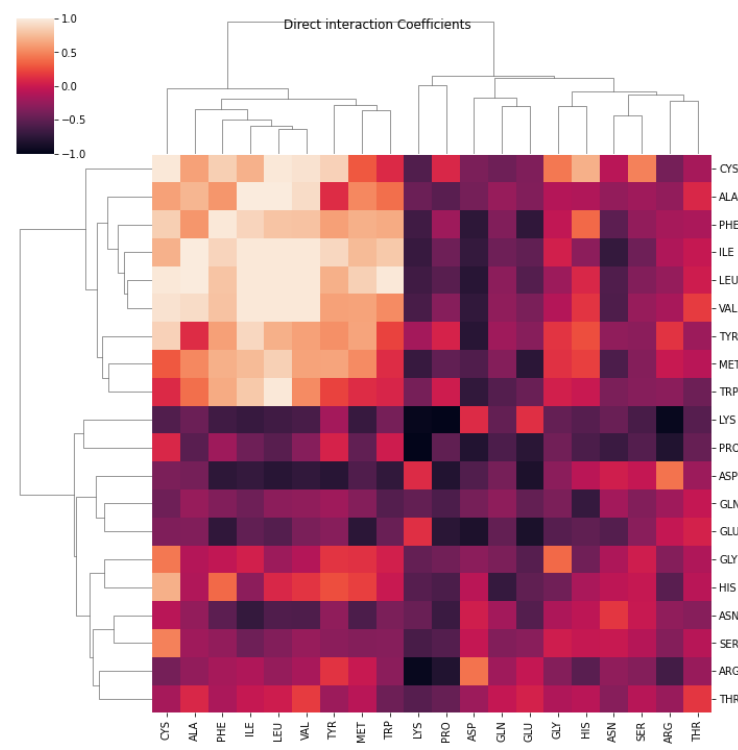


Assumption: Residues prefer to be in contact with other particular residues

$$V_{direct} = -\lambda_{direct} \sum_{j-i>9}^N \gamma_{ij}(a_i, a_j) \Theta_{ij}^I$$



- Range: From 4.5 to 6.5 Angstroms ($C_\beta - C_\beta$)



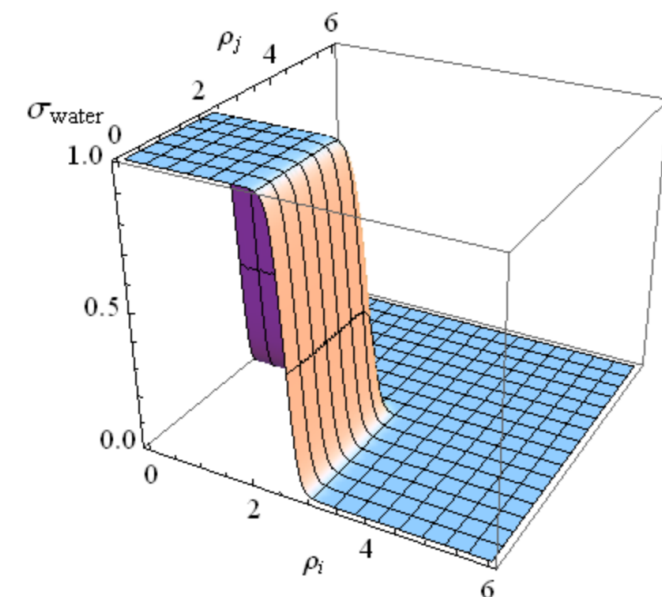
Mediated interactions

Assumption: Residues can be in contact through water-mediated interactions

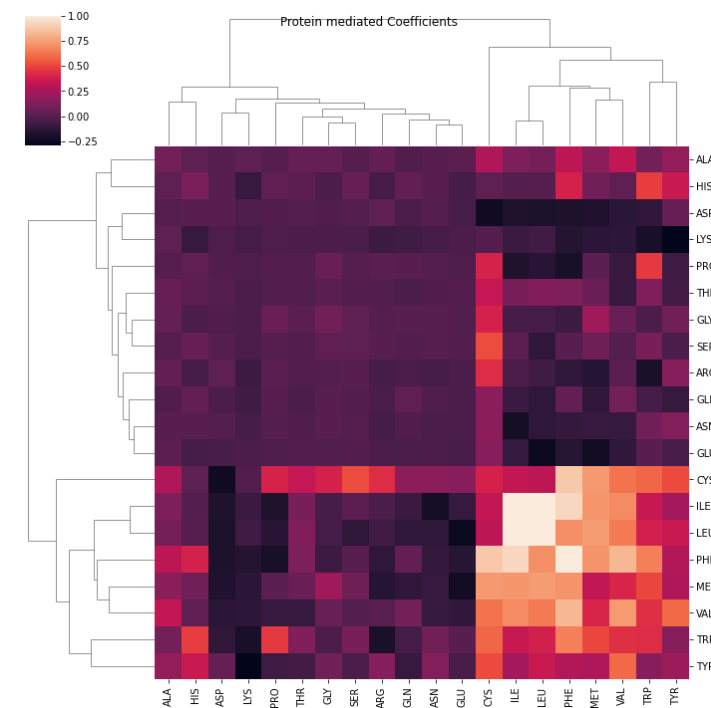
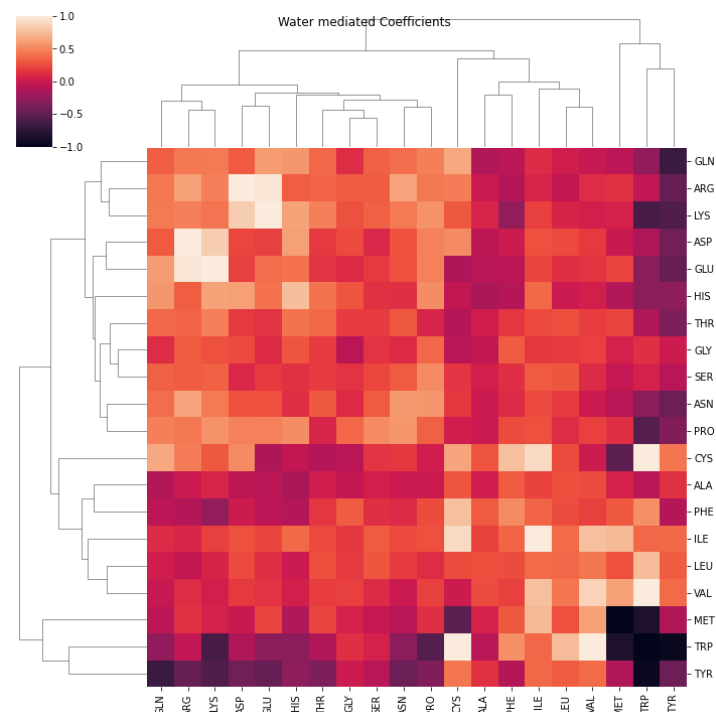
$$V_{\text{water-mediated}} = \lambda_{\text{water}} \sum_{j>i+n}^N \sum_i^N \Theta_{ij}^{II} \left(\sigma_{ij}^{\text{wat}} \gamma_{ij}^{\text{wat}}(a_i, a_j) \right)$$

$$V_{\text{protein-mediated}} = \lambda_{\text{prot}} \sum_{j>i+n}^N \sum_i^N \Theta_{ij}^{II} \left((1 - \sigma_{ij}^{\text{wat}}) \gamma_{ij}^{\text{prot}}(a_i, a_j) \right)$$

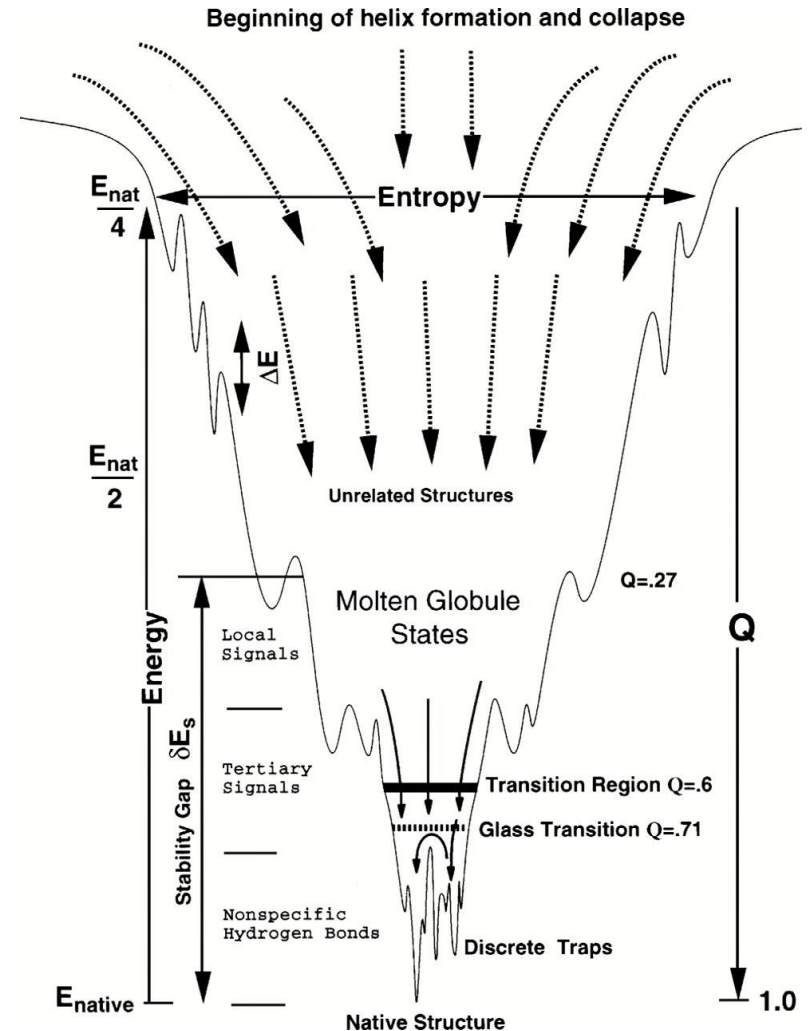
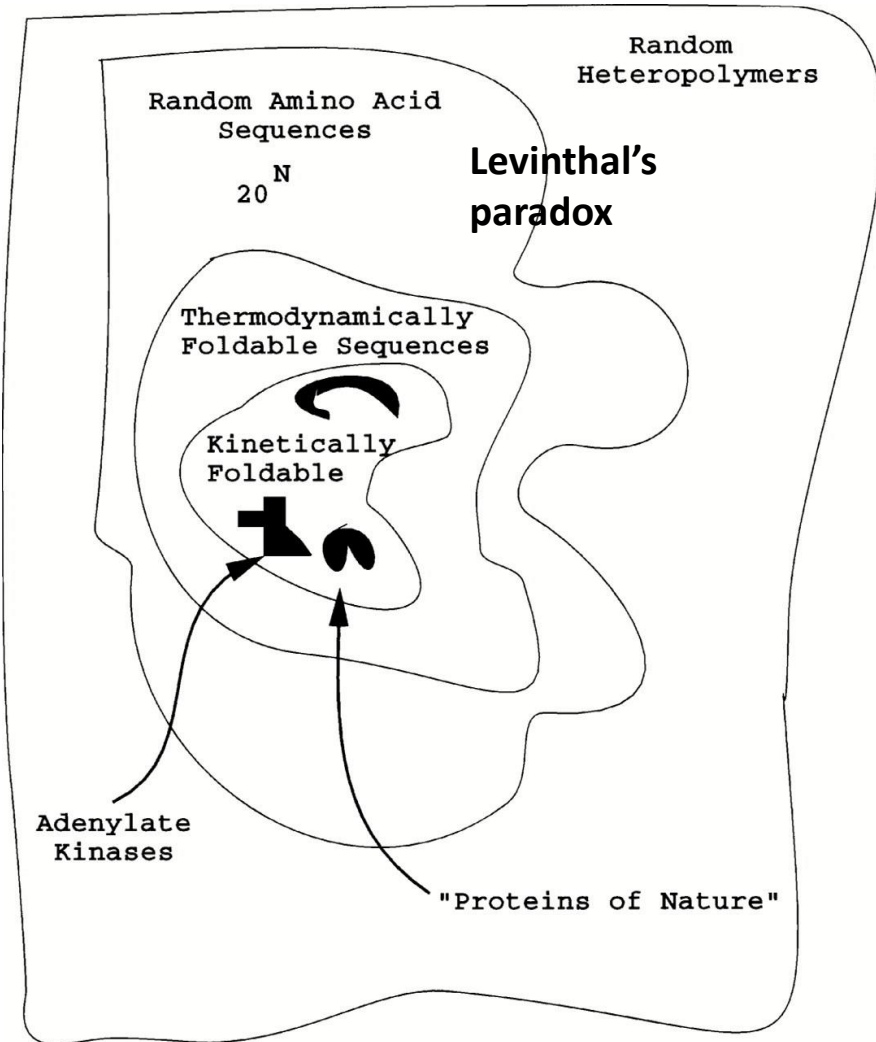
R R
OH₂
R R



Range: 6.5-9.5 Angstroms ($C_\beta - C_\beta$)



Parameter optimized using the principle of minimum frustration and the funneled energy landscape theory



Optimization of parameters

- Define Hamiltonian

$$E = \sum_i \gamma_i \varphi_i \quad \text{Contact and burial energy}$$

- Gather native and decoy structures, calculate A and B

$$A_i = \langle \varphi_i \rangle_{mg} - \varphi_n \quad \text{Decoy – native stability gap } (A\gamma)$$

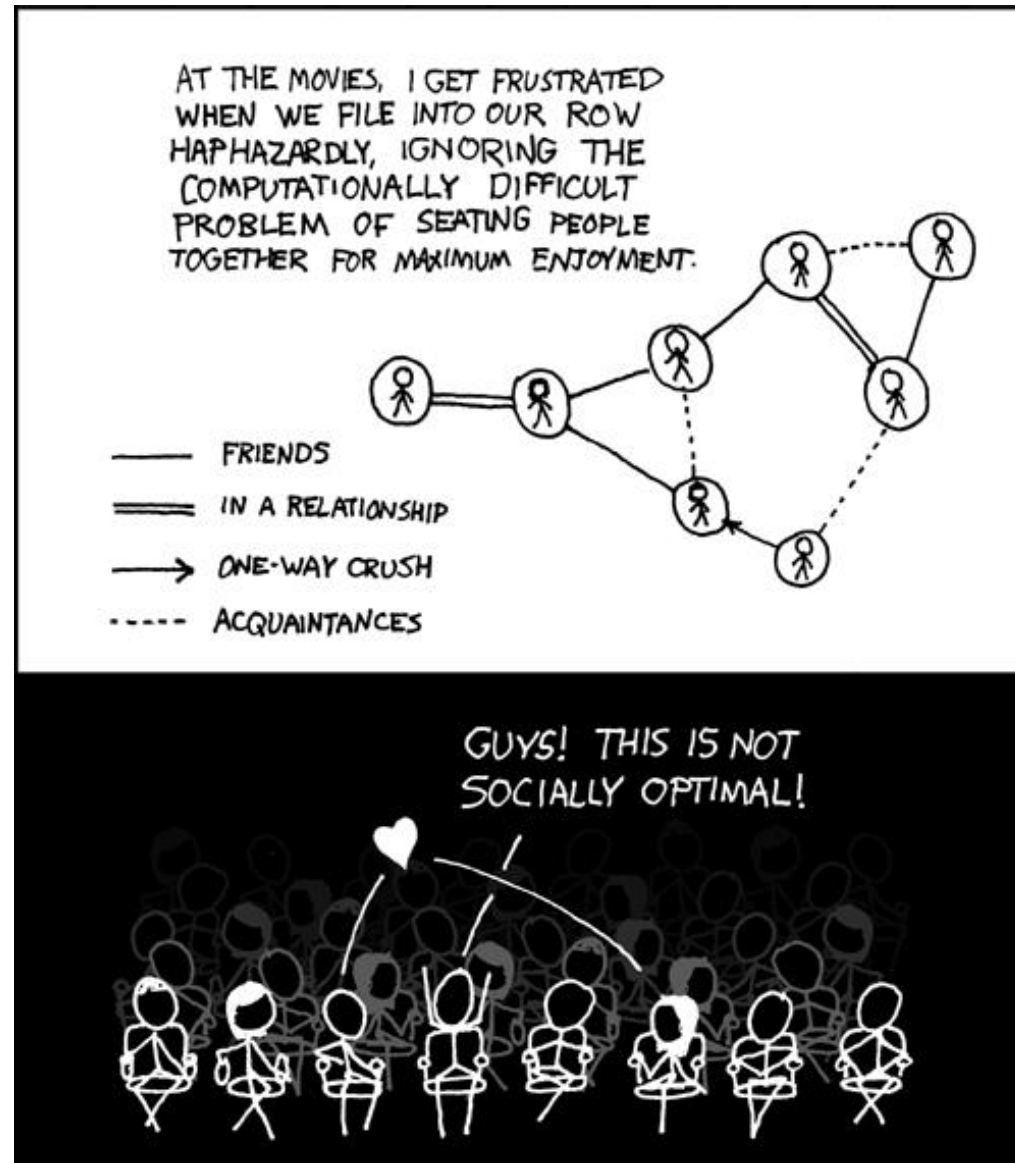
$$B_{ij} = \langle \varphi_i \varphi_j \rangle_{mg} - \langle \varphi_i \rangle_{mg} \langle \varphi_j \rangle_{mg} \quad \text{Decoy – native variance } (\gamma B \gamma)$$

- Get optimal gamma $\delta E_s / \Delta E = A\gamma / \sqrt{\gamma B \gamma}$

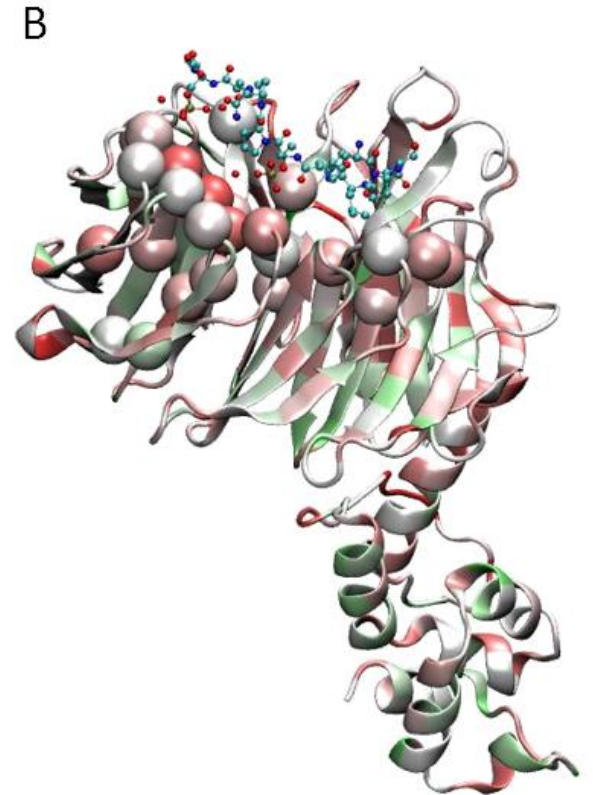
$$\gamma = B^{-1} A$$

- Generate new decoy structures and (Ideally) repeat until self-consistent

Frustration



xkcd.com/173



Frustration in proteins

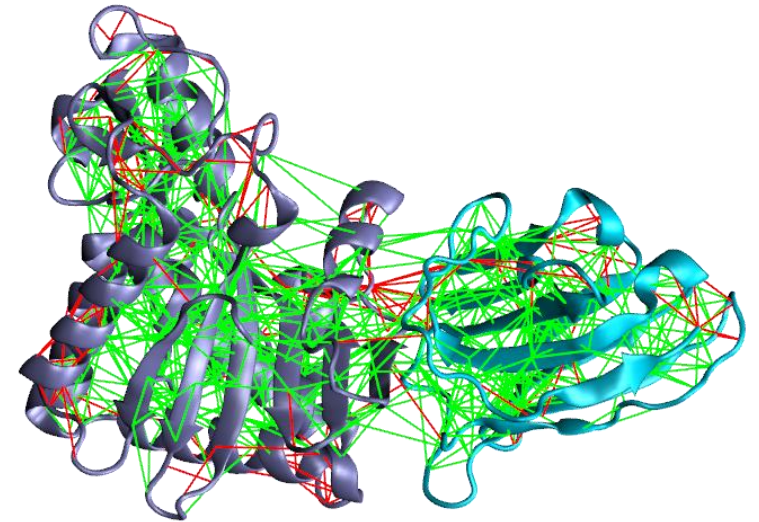
- Frustration is defined as a Z-score.
 - $F < -1$: Frustrated
 - $F > 0.78$: Minimally frustrated
- The energy only includes interactions.

$$F = \frac{E^N - \langle \mathbf{E}^D \rangle}{\sqrt{\frac{1}{N_D} \sum_{d=1}^{N_D} (E_d^D - \langle \mathbf{E}^D \rangle)^2}}$$

$$E^{ij} = E_{burial}^i + E_{burial}^j + E_{contact}^{ij} + E_{water}^{ij} + E_{electrostatic}^{ij}$$

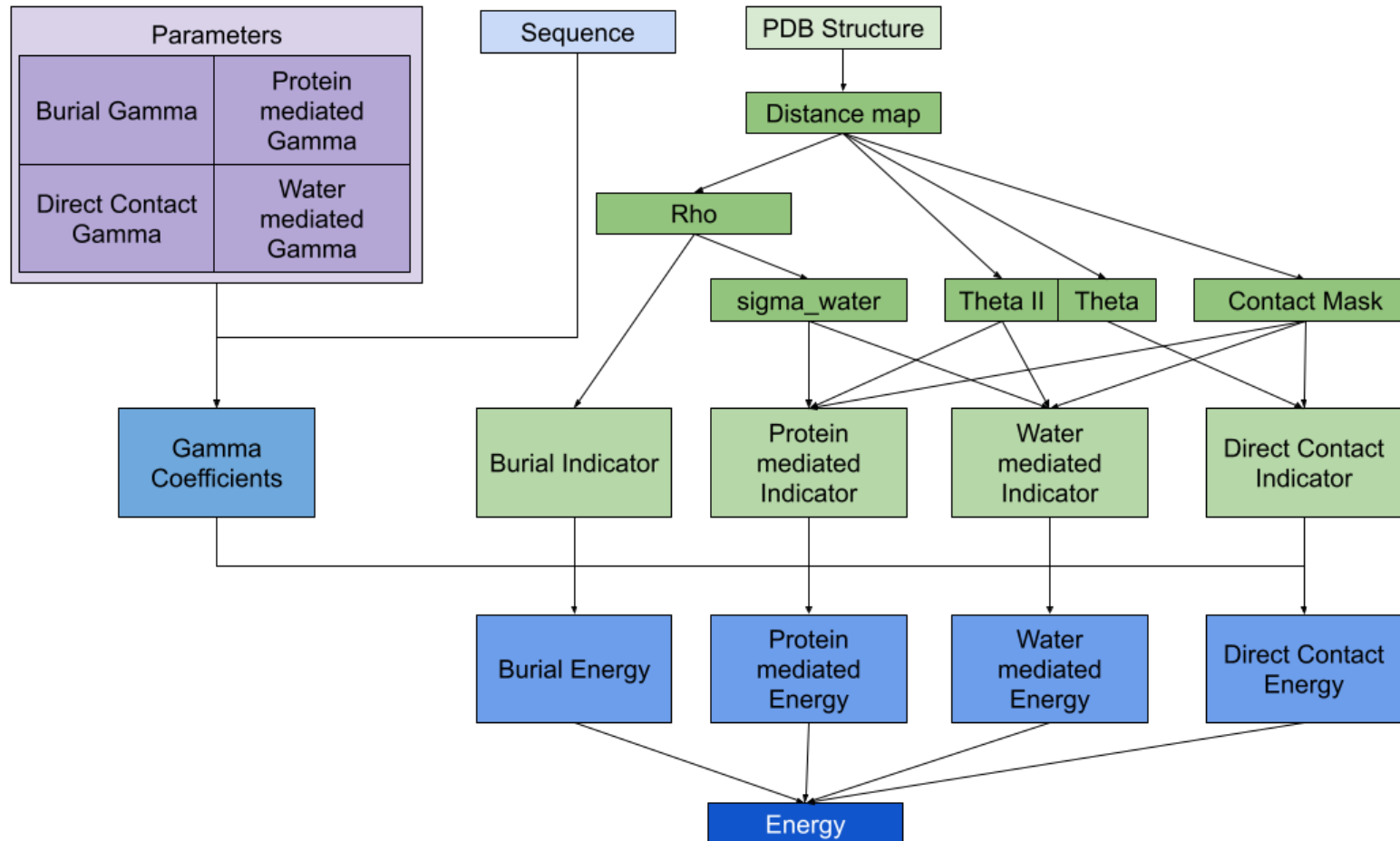
$$E_{electrostatic}^{ij} = K_{elec} \sum \frac{q_i q_j}{\epsilon_r r_{ij}} e^{\frac{-r_{ij}}{l_D}}$$

Energy of contact compared with a contact formed in the molten globule state



Overview of contact and burial potential calculations

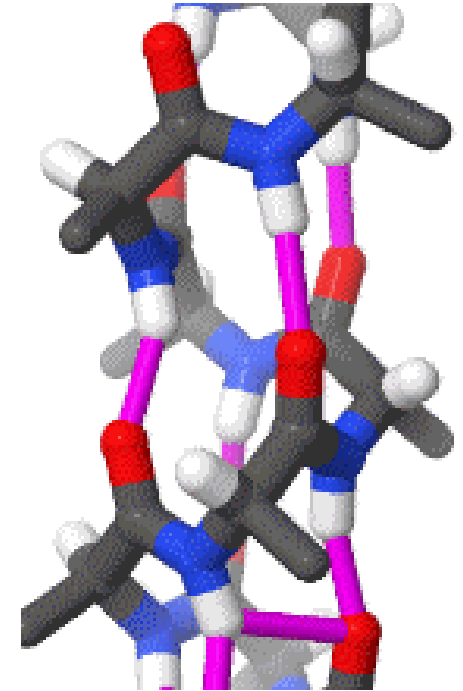
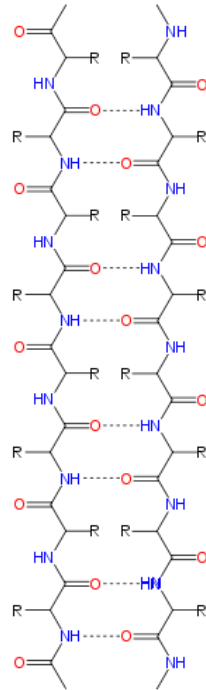
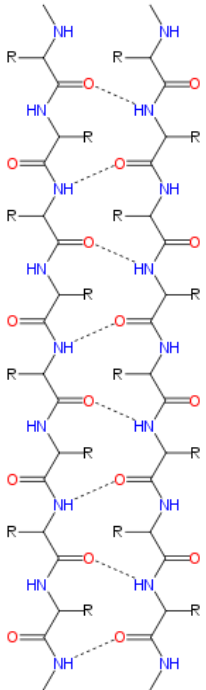
$$V_{contact} = V_{burial} + V_{direct} + V_{water-mediated} + V_{protein-mediated}$$



Hydrogen bonding terms

Assumption: Secondary structure forms as a cooperative process that depends on distance, orientation, and aminoacid nature

$$V_{HB} = V_{\beta} + V_{P-AP} + V_{helical}$$



Hydrogen bonding terms

$$V_{HB} = V_{\beta} + V_{P-AP} + V_{helical}$$

- Beta

$$V_{\beta}^{ij} = - \left[\Lambda_1(\underbrace{|j-i|}_{\text{sequence separation}}) \underbrace{\theta_{i,j}}_{\text{distance and orientation}} + \Lambda_2(\underbrace{a_i, a_j}_{\text{amino acid type}}, |j-i|) \theta_{i,j} \theta_{j,i} + \Lambda_3(a_i, a_j, |j-i|) \theta_{i,j} \theta_{j,i+2} \right] \underbrace{v_i^I v_j^{II}}_{\text{backbone extension}}$$

pairwise additive cooperative anti-parallel cooperative parallel

- P-AP

("Parallel, Anti-parallel")

Liquid crystal

$$V_{P-AP} = - \underbrace{\gamma_{APH}}_{\text{hairpin}} \sum_{i=1}^{N-13} \sum_{j=i+13}^{\min(i+16, N)} \underbrace{v_{i,j} v_{i+4, j-4}}_{\text{cooperative distances}} \\ - \underbrace{\gamma_{AP}}_{\text{anti-parallel}} \sum_{i=1}^{N-17} \sum_{j=i+17}^N v_{i,j} v_{i+4, j-4} - \underbrace{\gamma_P}_{\text{parallel}} \sum_{i=1}^{N-13} \sum_{j=i+9}^{N-4} v_{i,j} v_{i+4, j+4}$$

- Alpha

$$V_{helical} = - \lambda_{helical} \sum_{i=1}^{N-4} \underbrace{(f(a_i) + f(a_{i+4}))}_{\text{helical probability}} \left(\gamma_{prot} \underbrace{\sigma_{i,i+4}^{prot}}_{\text{buried}} + \gamma_{wat} \underbrace{\sigma_{i,i+4}^{wat}}_{\text{exposed}} \right) \times \\ \exp \left[\underbrace{- \frac{(r_{i,i+4}^{ON} - \langle r^{ON} \rangle)^2}{2\sigma_{ON}^2} - \frac{(r_{i,i+4}^{OH} - \langle r^{OH} \rangle)^2}{2\sigma_{OH}^2}}_{\text{distance and orientation}} \right]$$

Associative memory term

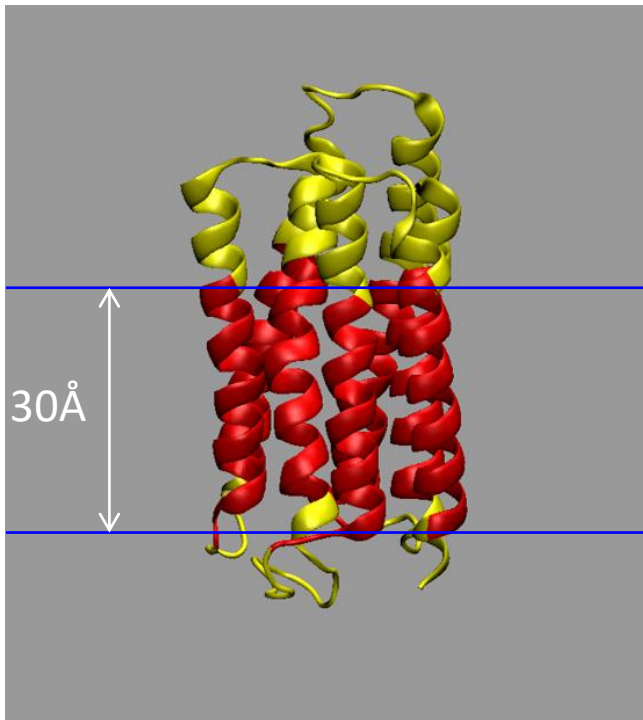
Assumption: The details absent from the AWSEM model can be extracted from known structures.

$$V_{AM} = -\lambda_{AM} \sum_m \omega_m \sum_{ij} \gamma_{ij} \exp \left[-\frac{(r_{ij} - r_{ij}^m)^2}{2\sigma_{IJ}^2} \right]$$

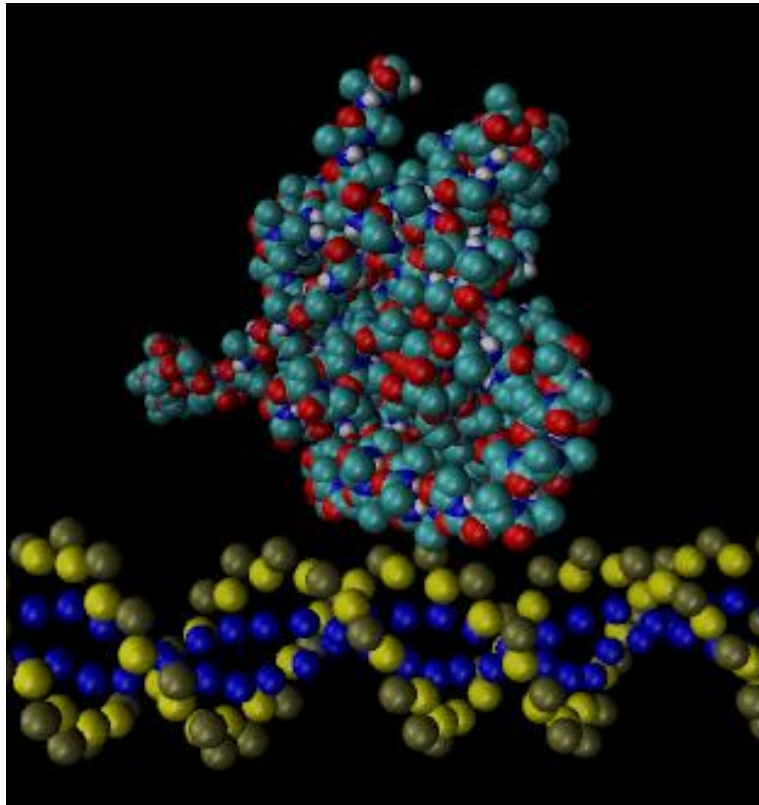
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- Parameters: Memories can be obtained by using bioinformatic searches to obtain global or local alignments or by simulating fragments using all-atom simulations

Extra terms

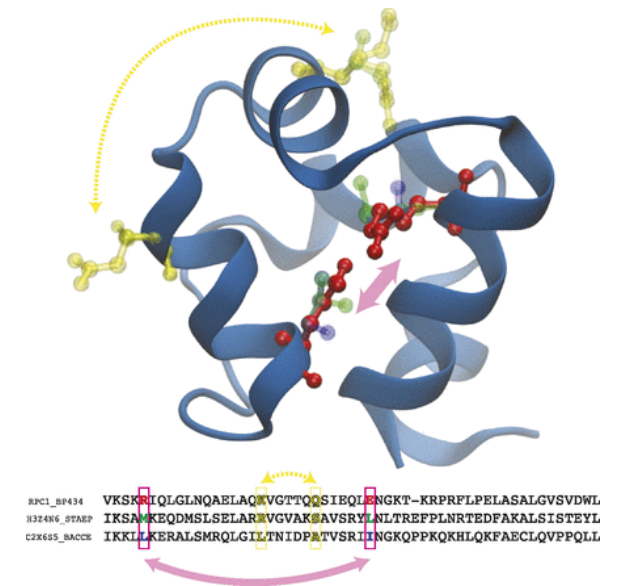
Membrane Potential



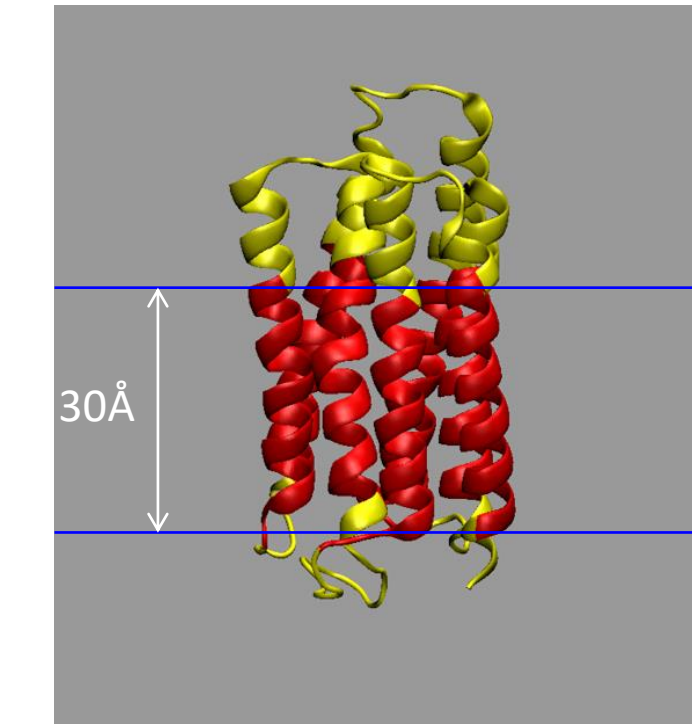
Open3SPN2 DNA model



Evolutionary restraints



Implicit membrane term



Residues in **cytoplasmic** and **periplasmic** environments are treated equally.

$$V_{Membrane} = V_{Periplasmic} d_{zpred,periplasmic} + V_{Cytoplasmic} d_{zpred,cytoplasmic} + V_{Transmembrane} d_{zpred,transmembrane}$$

Each potential term only influences the residues assigned to be in their respective environments.

In addition to the above z-dependent potential, the membrane protein model includes a reoptimized contact term (based on a database of membrane protein structures) and optionally includes terms that mimic pore-formation in the membrane and lateral pressure/liquid-crystal ordering of the transmembrane helices.

Implicit membrane term membrane gammas

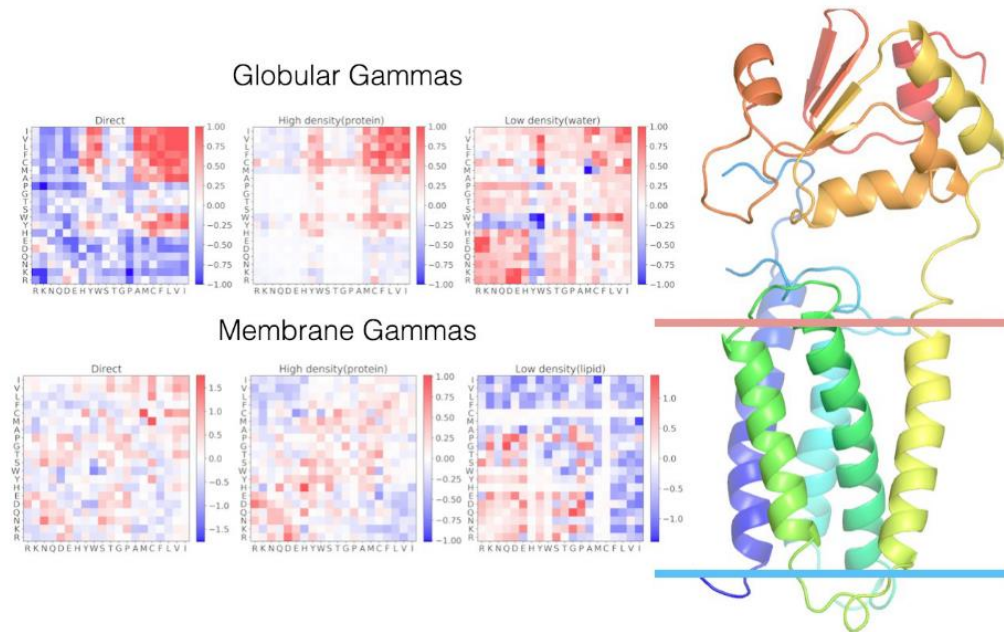
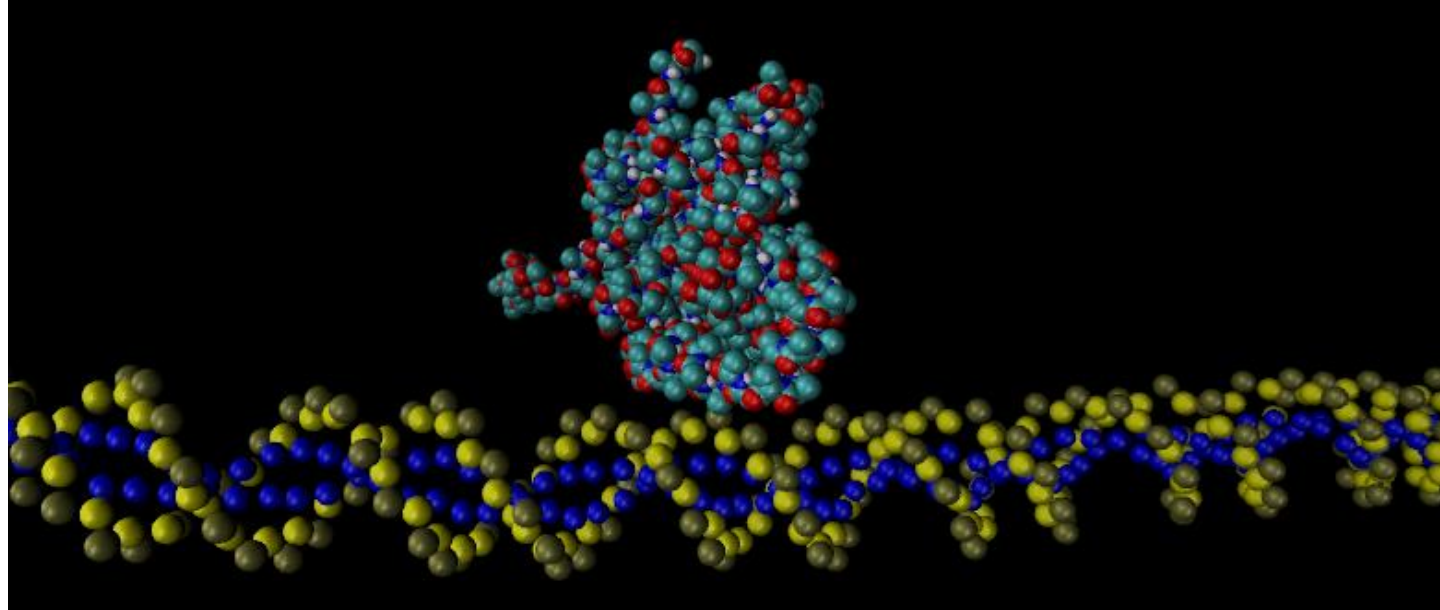


Fig 5. A schematic figure for the Z-dependent contact potential. The residues outside of the membrane, where the membrane boundary is indicated by the two colored lines, interact using the globular parameters. The residues inside the membrane interact using the membrane-optimized parameters. If one residue is inside, while another one is outside, the pair interacts as if they both were in water. In the heat maps on the left side of the figure, red color indicates a favorable interaction between the pair of residues indicated on the horizontal and vertical axes, whereas blue color indicates an unfavorable interaction. Separate heat maps are shown for the direct, low-density, and high-density interaction matrices in the water (globular) and membrane environments.

Interactions with DNA (open3SPN2)



$$U_{3SPN2-AWSEM} = U_{Exclusion} + U_{Electrostatics}$$

$$U_{Exclusion} = \sum \epsilon_r \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \epsilon_r \quad r < \sigma_{ij}$$

$$V_{DH} = K_{Elec} \sum_{i < j} \frac{q_i q_j}{\epsilon_r r_{ij}} e^{-r_{ij}/l_D}$$

Lu W, Bueno C, Schafer NP, Moller J, Jin S, Chen X, et al. OpenAWSEM with Open3SPN2: A fast, flexible, and accessible framework for large-scale coarse-grained biomolecular simulations. PLoS Comput Biol. 2021;17: e1008308.
doi:10.1371/journal.pcbi.1008308. PubMed PMID: 33577557

3SPN2 potential

$$U_{3SPN2} = U_{Bond} + U_{Angle} + U_{Dihedral} + U_{Stacking} + U_{BasePair} + U_{CrossStacking} + U_{Exclusion} + U_{Electrostatics}$$

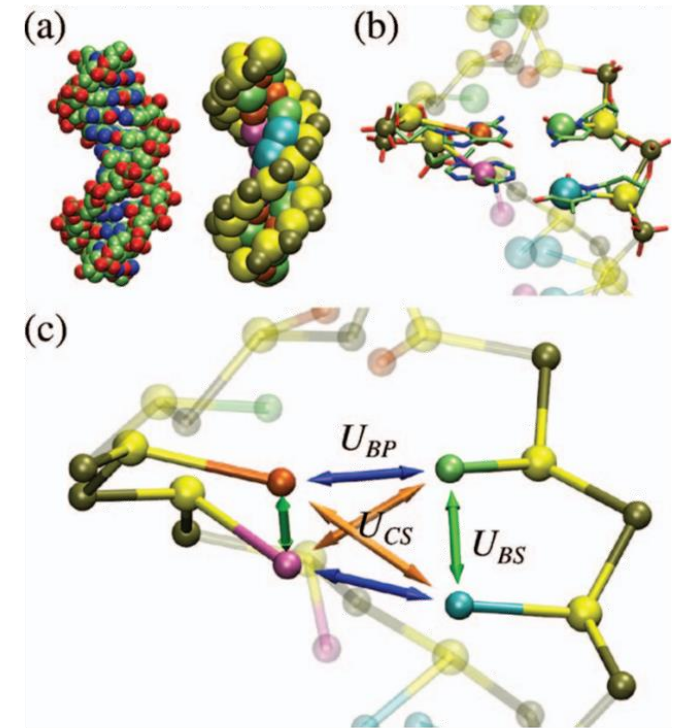
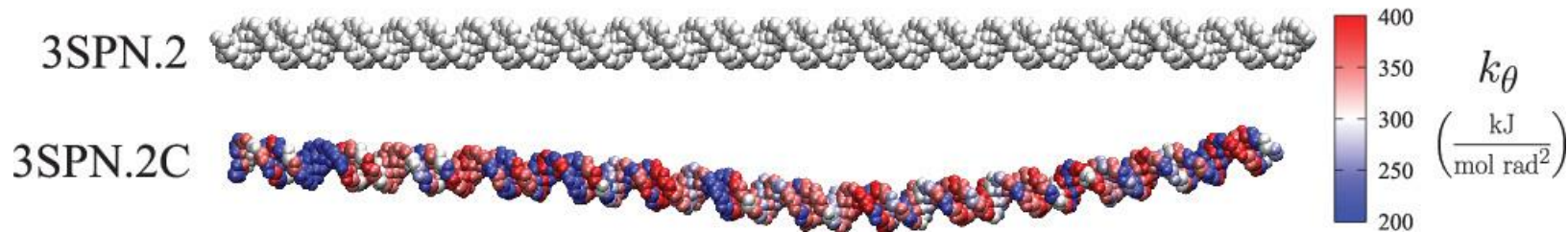


FIG. 1. (a) Comparison of all-atom and 3SPN.2 coarse-grained excluded volume representations of the Drew-Dickerson dodecamer (5'-CGCGAATTCGCG-3'). (b) Coarse-grained sites of 3SPN.2 superimposed on the all-atom representations. The coarse-grained sites are located at the centers of mass of the phosphate, sugar, or base. (c) Schematic representation of the angle-dependent non-bonded interactions acting between the base sites. The green arrows represent the base stacking potential U_{BS} , the blue arrows represent the base pairing potential U_{BP} , and the orange arrows represent the cross-stacking potential U_{CS} . Figures were rendered using VMD.⁵⁰

Parametrization of the 3SPN2 forcefield

- Free energy of hybridization (T_m)
- Intra-strand base stacking (Stack/Unstacked conformational transition)
- Persistence length
- Width of minor and major grooves

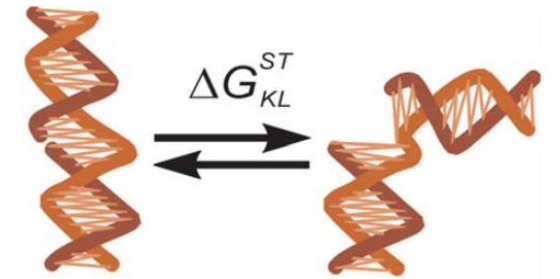


Figure 1. Schematic representation of the stacked-to-unstacked conformational transition of the DNA fragment at the nick site described by the energy difference, ΔG_{KL}^{ST} .

Hinckley, D. M., Freeman, G. S., Whitmer, J. K., & De Pablo, J. J. (2013). An experimentally-informed coarse-grained 3-site-per-nucleotide model of DNA: Structure, thermodynamics, and dynamics of hybridization. *Journal of Chemical Physics*, 139(14). <https://doi.org/10.1063/1.4822042>

Protozanova, E., Yakovchuk, P., & Frank-Kamenetskii, M. D. (2004). Stacked-unstacked equilibrium at the nick site of DNA. *Journal of Molecular Biology*, 342(3), 775–785. <https://doi.org/10.1016/j.jmb.2004.07.075>

AWSEM-ER <-Direct coupling analysis (DCA)

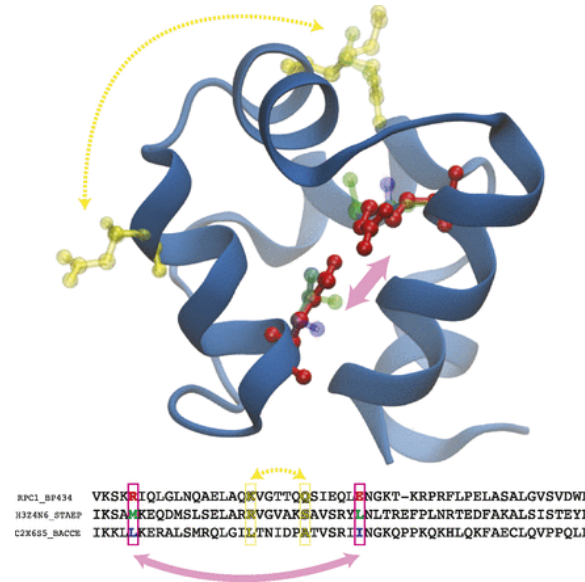
$$H(\sigma) = - \sum_{i=1}^L \boxed{\text{Fields}} h_i(\sigma_i) - \sum_{i=1}^{L-1} \sum_{j=i+1}^L \boxed{\text{Couplings}} J_{i,j}(\sigma_i, \sigma_j)$$

$$P(\sigma) = \frac{1}{Z} e^{-\frac{H(\sigma)}{k_b T_{sel}}}$$

$$H(\sigma) = - \sum_{i=1}^{L-1} \sum_{j=i+1}^L J_{i,j}(\sigma_i, \sigma_j) \boxed{\text{Distance cutoff}} \Theta(d - r_{i,j}) - \sum_{i=1}^L h_i(\sigma_i)$$

$$\Delta H(\sigma_1, \sigma_2) = H(\sigma_2) - H(\sigma_1)$$

$$V_{ER} = -\frac{1}{2} \sum_i \left| \sum_j^{r_{ij} < r_c} -\gamma_{ij} \left| \frac{e}{a} \right|^p e^{-\frac{(r_{ij} - r_{ij}^N)^2}{2\sigma_{ij}^2}} \right|^p$$



Morcos, Faruck, Andrea Pagnani, Bryan Lunt, Arianna Bertolino, Debora S. Marks, Chris Sander, Riccardo Zecchina, José N. Onuchic, Terence Hwa, and Martin Weigt. 2011. "Direct-Coupling Analysis of Residue Coevolution Captures Native Contacts across Many Protein Families." *PNAS* 108 (49):E1293–1301.

<https://doi.org/10.1073/pnas.11114711108>.

Ekeberg, M., Lövkvist, C., Lan, Y., Weigt, M., & Aurell, E. (2013). Improved contact prediction in proteins: Using pseudolikelihoods to infer Potts models. *Physical Review E*, 87(1), 1–19. <http://doi.org/10.1103/PhysRevE.87.012707>

Morcos, F., Schafer, N. P., Cheng, R. R., Onuchic, J. N., & Wolynes, P. G. (2014). Coevolutionary information, protein folding landscapes, and the thermodynamics of natural selection. *PNAS*, 111(34), 12408–12413. <http://doi.org/10.1073/pnas.1413575111>

Sirovetz BJ, Schafer NP, Wolynes PG. Protein Structure Prediction: Making AWSEM AWSEM-ER by Adding Evolutionary Restraints. *Proteins*. 2017;85: 2127–2142. doi:10.1002/prot.25367. PubMed PMID: 28799172

Other terms

- Electrostatics

$$V_{\text{DH}} = K_{\text{Elec}} \sum_{i < j} \frac{q_i q_j}{\epsilon_r r_{ij}} e^{-r_{ij}/l_D}$$

- Desolvation $V_{\text{DSB}} = \lambda_{\text{DSB}} \sum_{j-i>9}^N \frac{1}{2} \left(\tanh \left[\kappa_{\text{DSB}} (r_{ij} - r_{\min}^{\text{DSB}}(a_i, a_j)) \right] + \right.$

- Go-term $\left. \tanh \left[\kappa_{\text{DSB}} (r_{\max}^{\text{DSB}}(a_i, a_j) - r_{ij}) \right] \right)$

$$V_{\text{Go}} = -\frac{1}{2} \sum_i \left| \sum_j^{r_{ij} < r_c} -\gamma_{ij} \left| \frac{e}{a} \right|^p e^{-\frac{(r_{ij} - r_{ij}^N)^2}{2\sigma_{ij}^2}} \right|$$

The implementation of AWSEM and 3SPN2 in openMM eases the development of new potential terms

Example: Transient disulfide bonds

$$V_{disulfide} = k \sum V_{ij}$$

$$V_{ij} = \alpha(r_{ij}) \theta_{ij}^{near} \theta_{ij}^{small}$$

$$\alpha(r_{ij}) = \frac{1}{2} \tanh(\kappa(r_{ij} - r_c)) - \frac{1}{2}$$

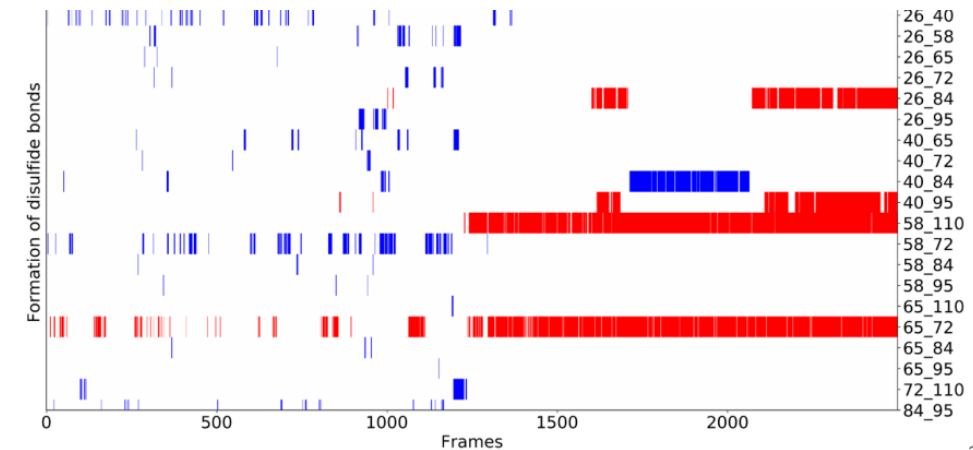
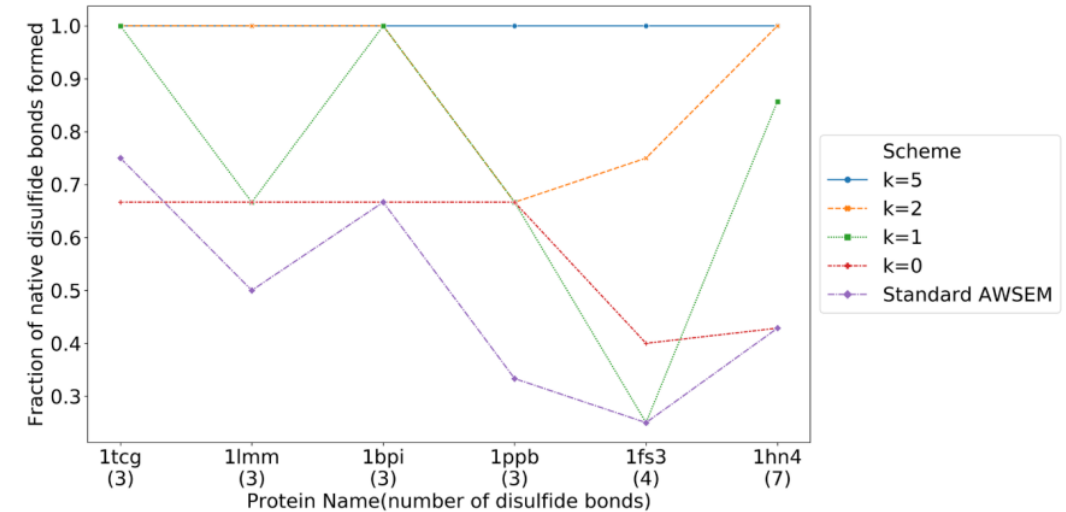
$$\theta_{ij}^{near} = \frac{1}{2} \tanh\left(\kappa_s \left(0.2 - \left|\rho_i^{cys} - \rho_j^{cys}\right|\right)\right) + \frac{1}{2}$$

$$\theta_{ij}^{small} = \frac{1}{2} \tanh\left(\kappa_s \left(2.2 - \left(\rho_i^{cys} + \rho_j^{cys}\right)\right)\right) + \frac{1}{2}$$

$$\rho_i^{cys} = \sum_{|j-i|>1} \frac{1}{2} (1 - \tanh(\kappa(r - r_c)))$$

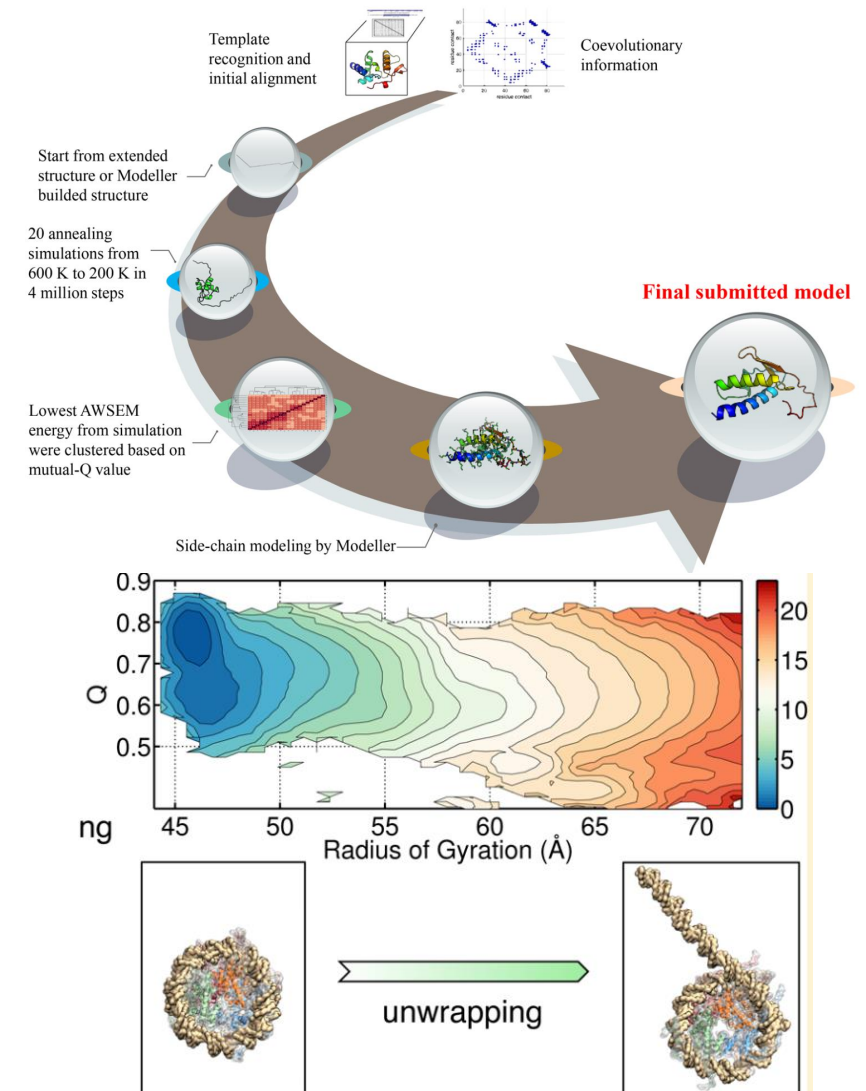
Red: Native contacts

Cyan: Non-native contacts

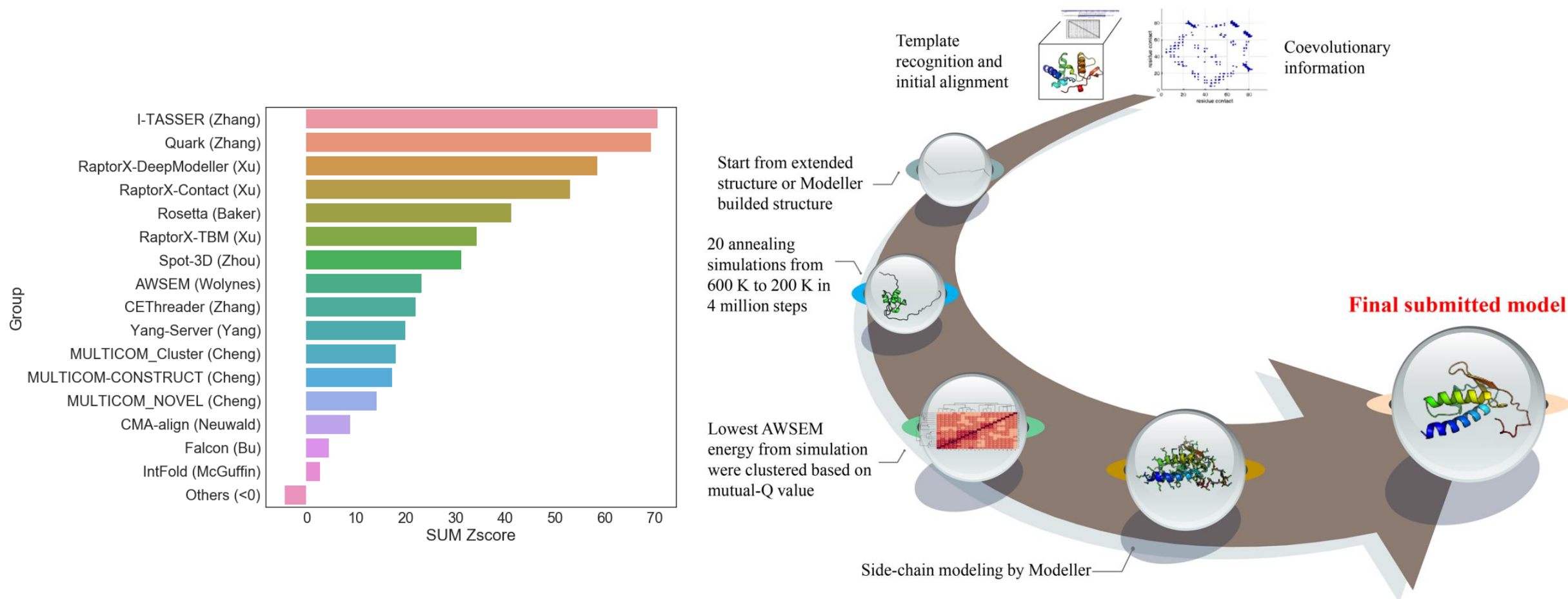


Some applications

- Protein structure prediction.
- Protein folding and misfolding pathways.
- Protein binding and aggregation
- Association with DNA, RNA and membranes.
- Frustration and stability prediction.



Protein Structure prediction (CASP13)



Structure Prediction

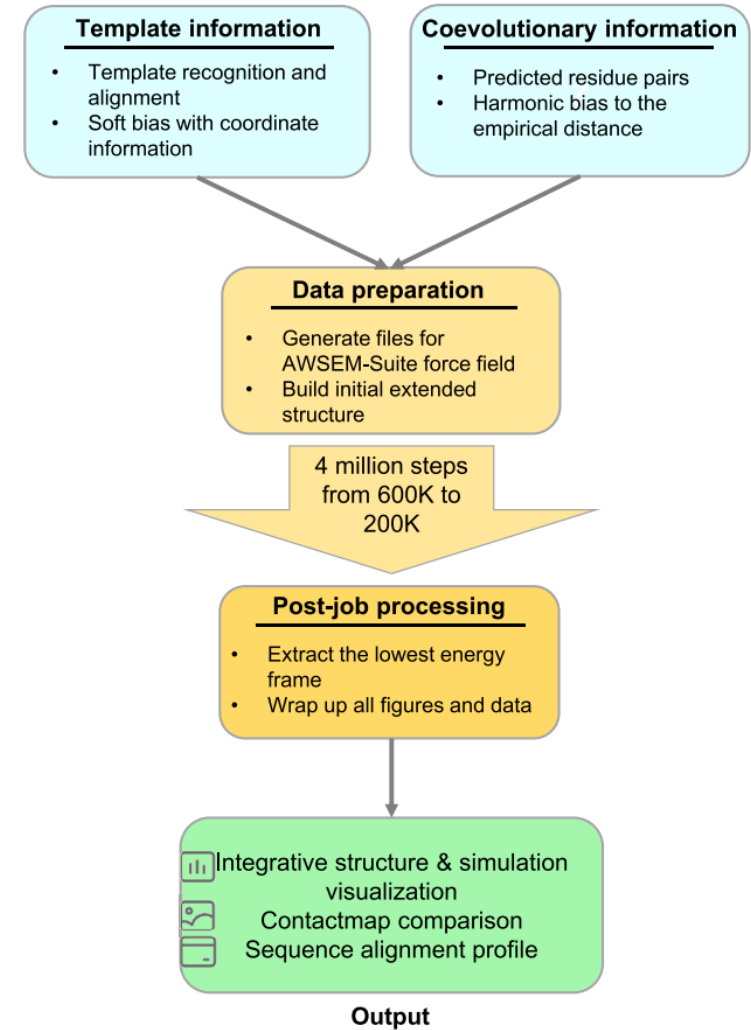
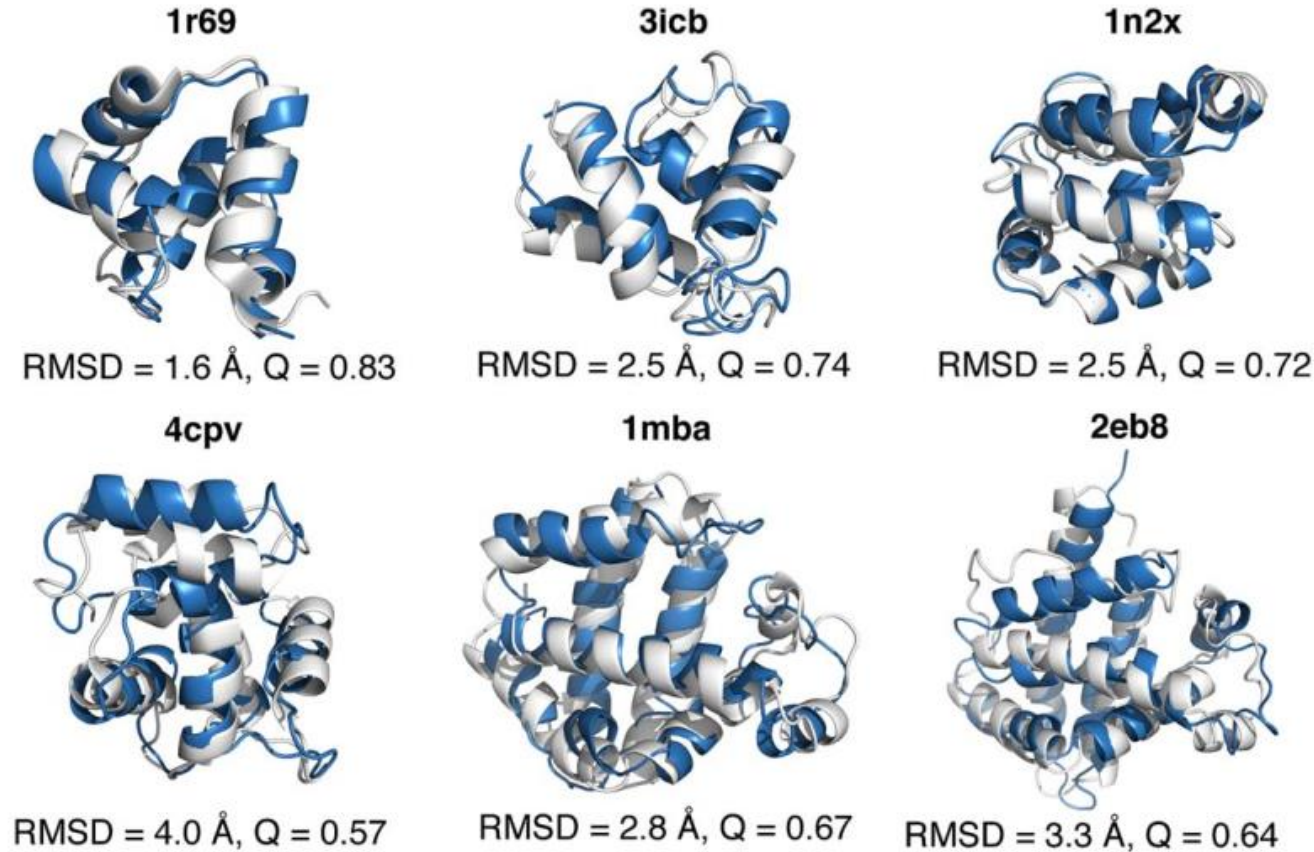
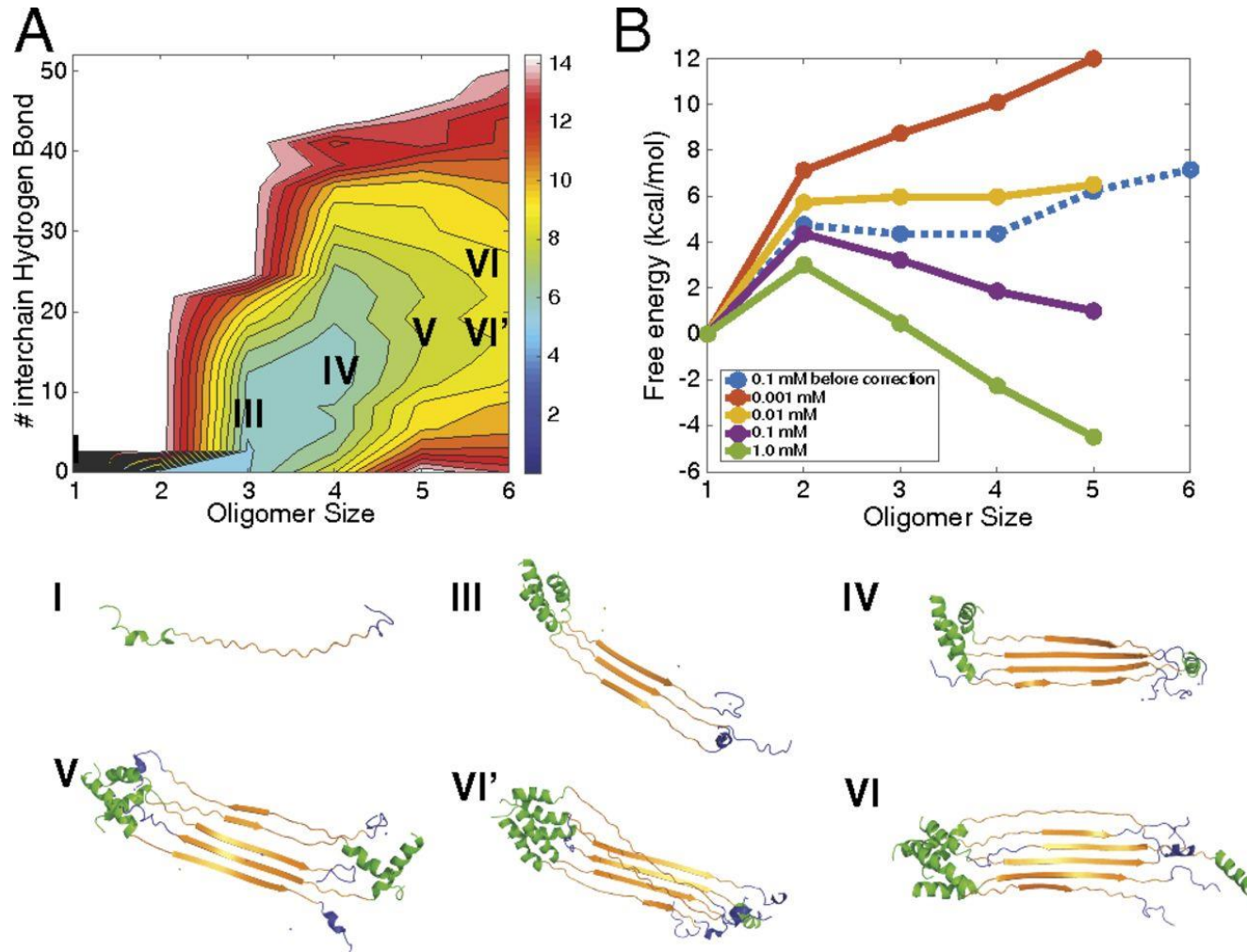


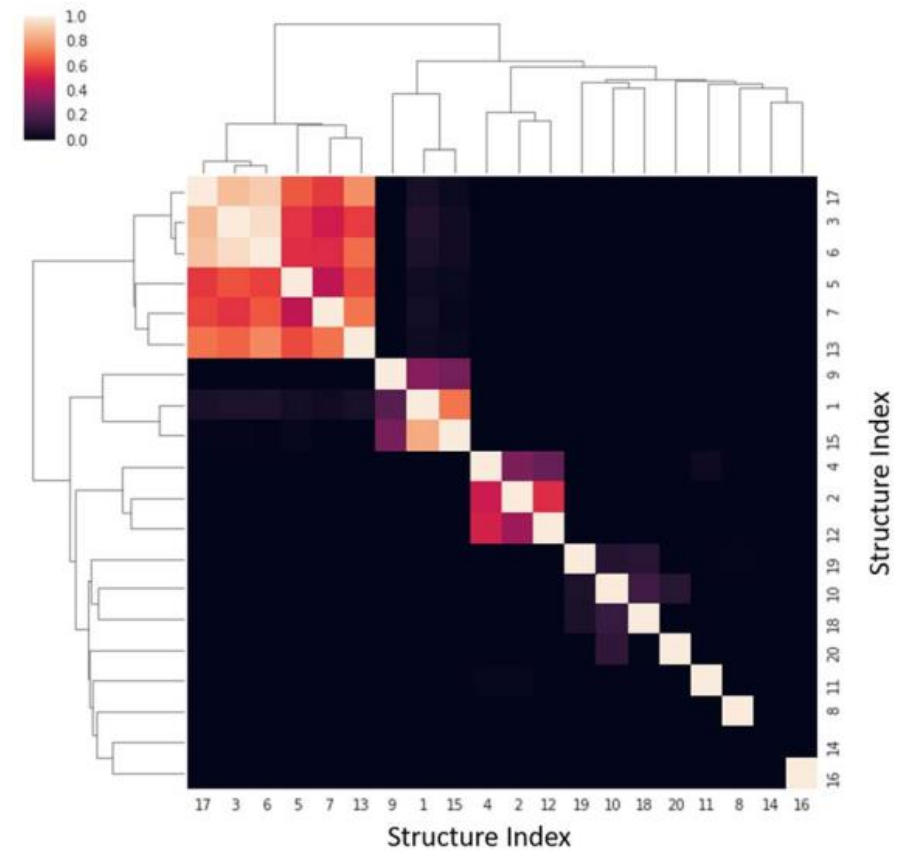
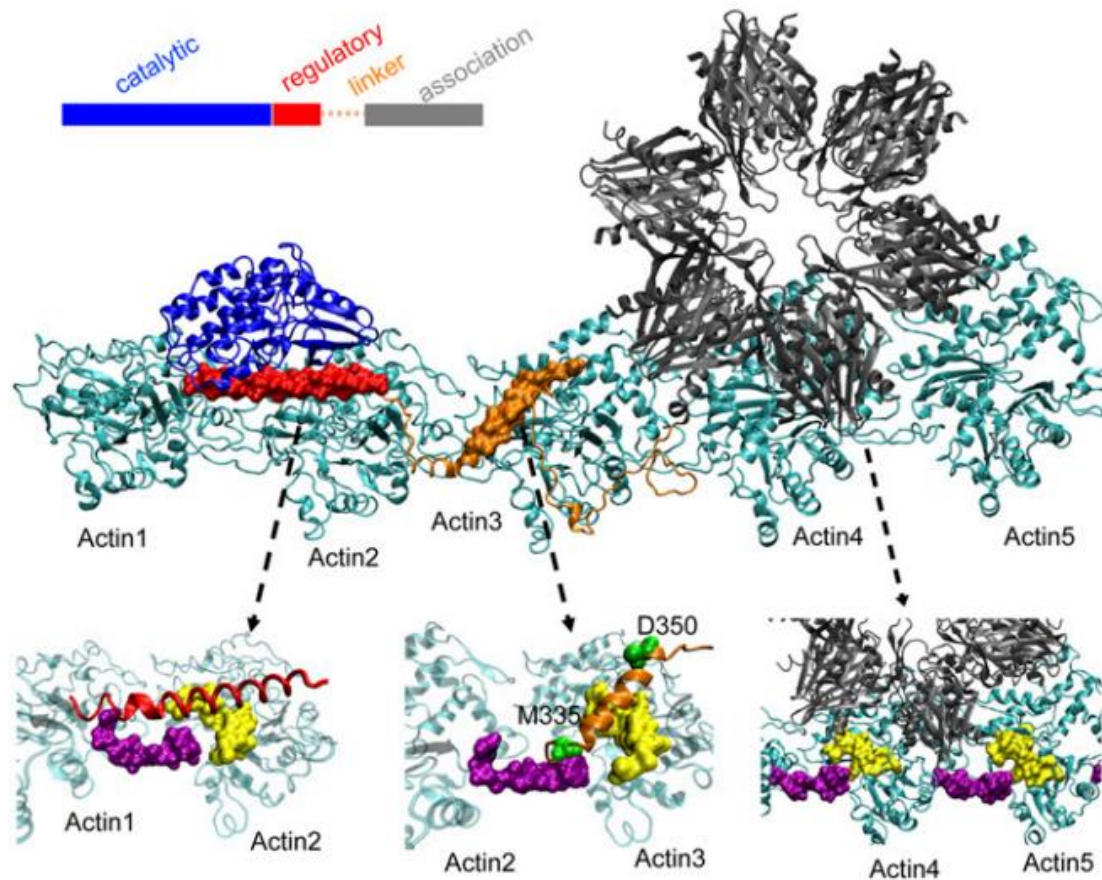
Figure 1. Flowchart of the AWSEM-Suite protocol.

Protein aggregation and misfolding pathways



Chen, Mingchen, and Peter G. Wolynes. 2017. "Aggregation Landscapes of Huntingtin Exon 1 Protein Fragments and the Critical Repeat Length for the Onset of Huntington's Disease." *Proceedings of the National Academy of Sciences of the United States of America* 114 (17): 4406–11. <https://doi.org/10.1073/pnas.1702237114>.

Protein Docking



Transmembrane protein folding pathways

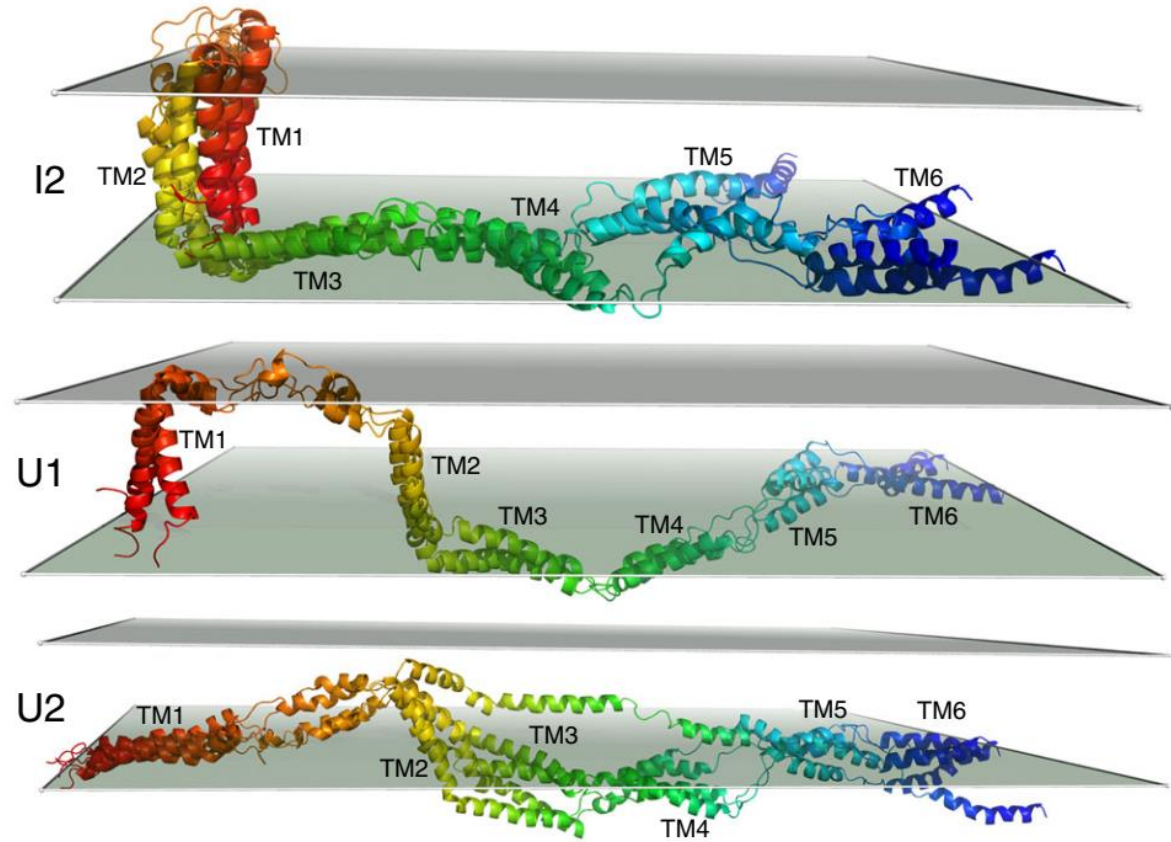
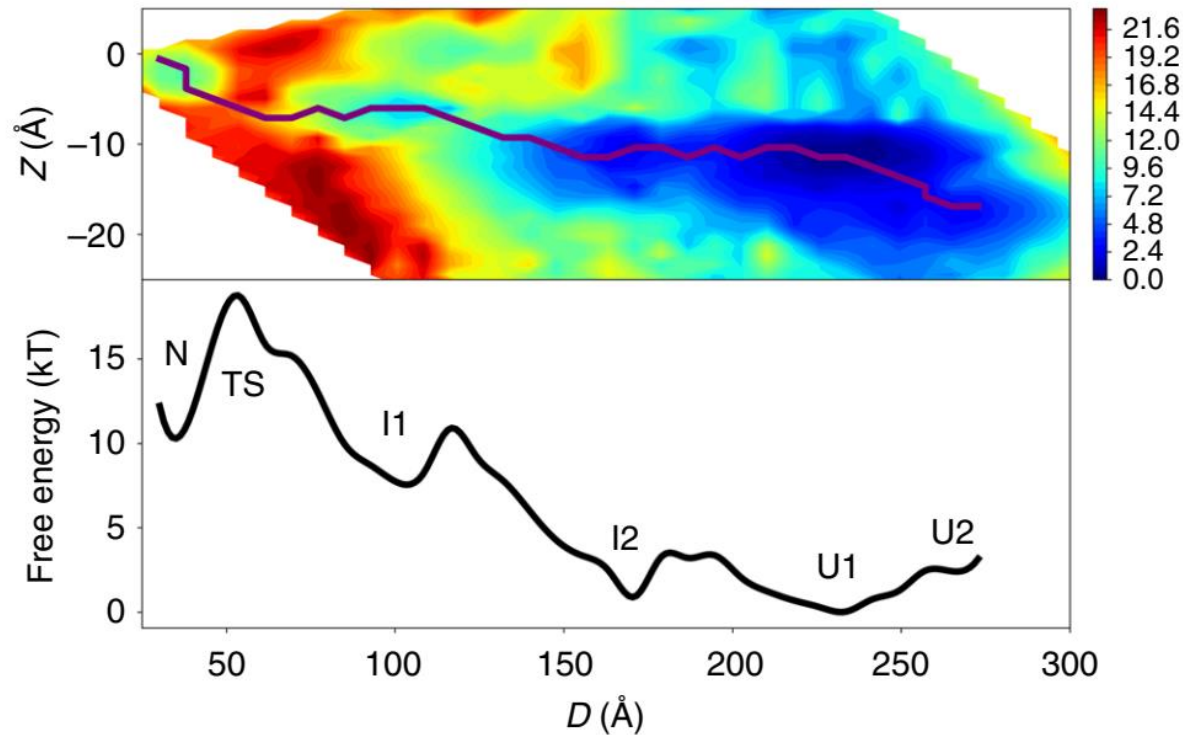
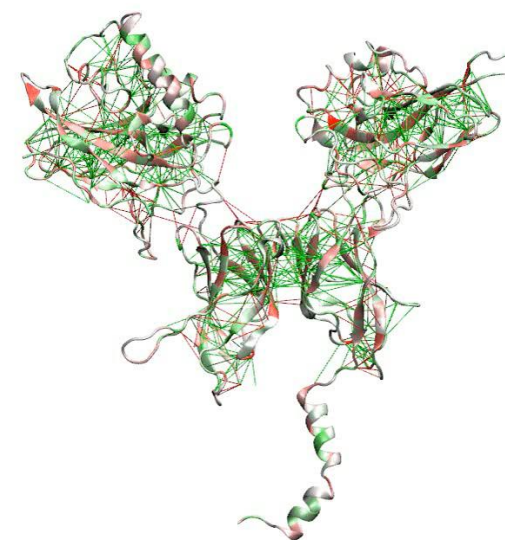
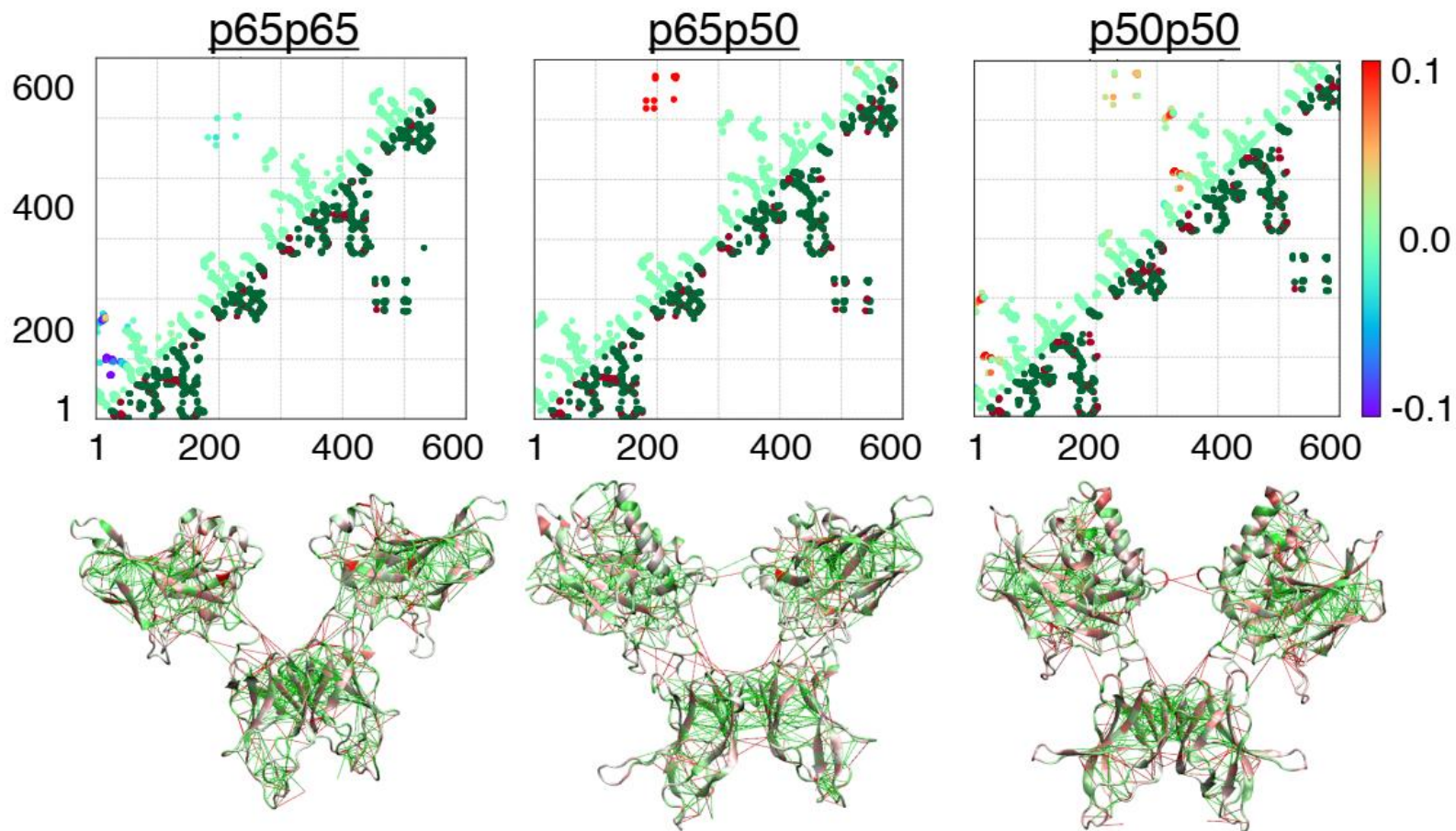
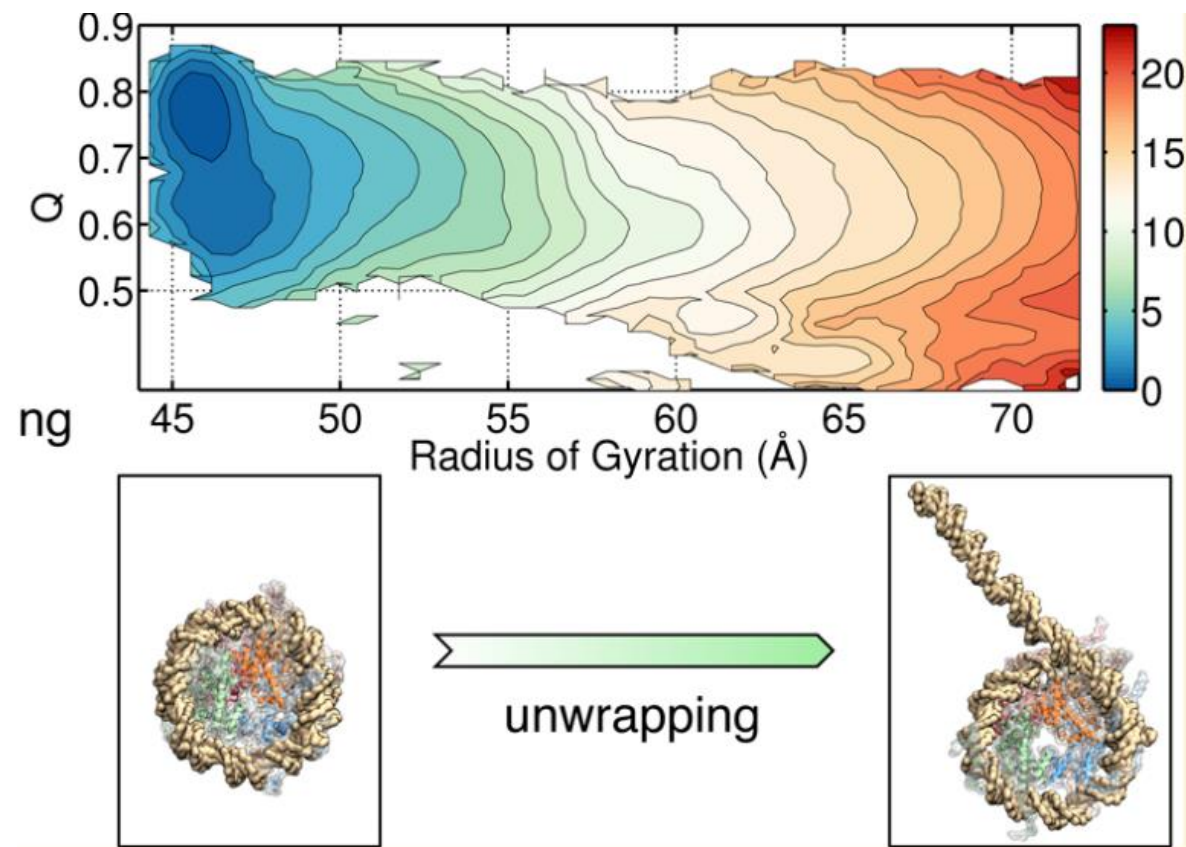
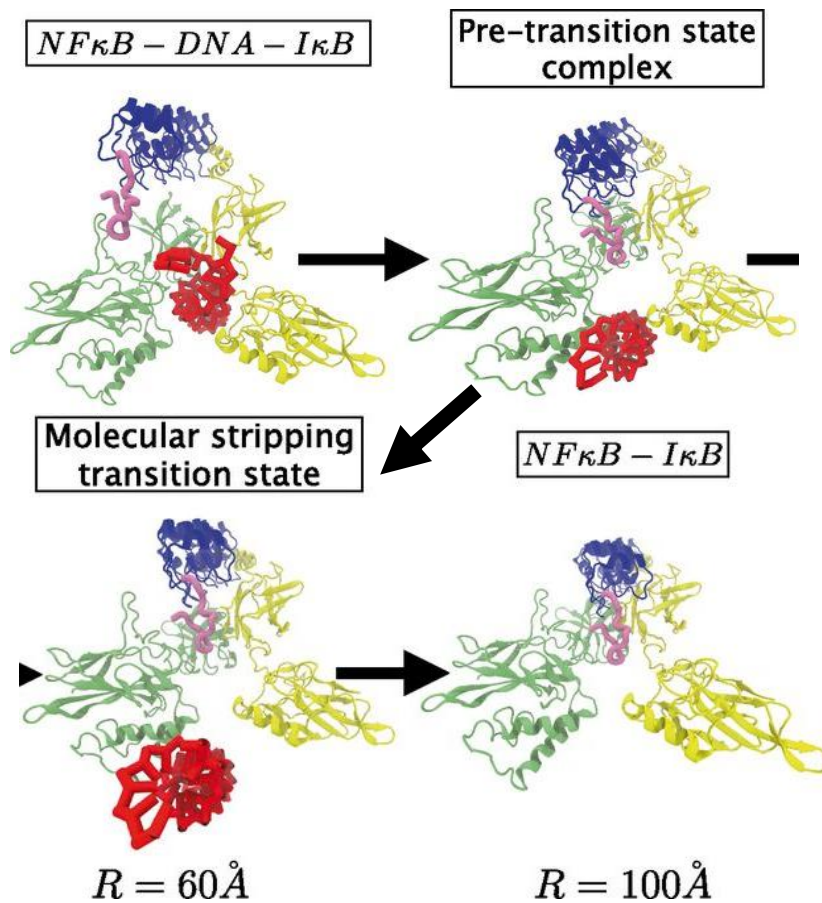


Fig. 3 Free-energy landscape underlying GlpG's unfolding and extraction from a bilayer under high applied force (Top) Two-dimensional free energy

Protein dynamics



Protein-DNA Interactions



Morozov, A. V., Fortney, K., Gaykalova, D. A., Studitsky, V. M., Widom, J., & Siggia, E. D. (2009). Using DNA mechanics to predict in vitro nucleosome positions and formation energies. *Nucleic Acids Research*, 37(14), 4707–4722. <https://doi.org/10.1093/nar/gkp475>

Zhang, B., Zheng, W., Papoian, G. A., & Wolynes, P. G. (2016). Exploring the Free Energy Landscape of Nucleosomes. *Journal of the American Chemical Society*, 138(26), 8126–8133. <https://doi.org/10.1021/jacs.6b02893>

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- Jin S, Chen M, Chen X, Bueno C, Lu W, Schafer NP, et al. Protein Structure Prediction in CASP13 Using AWSEM-Suite. J Chem Theory Comput. 2020;16: 3977–3988. doi:10.1021/acs.jctc.0c00188. PubMed PMID: 32396727 **(Structure Prediction protocol)**
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